

BFS Traversal

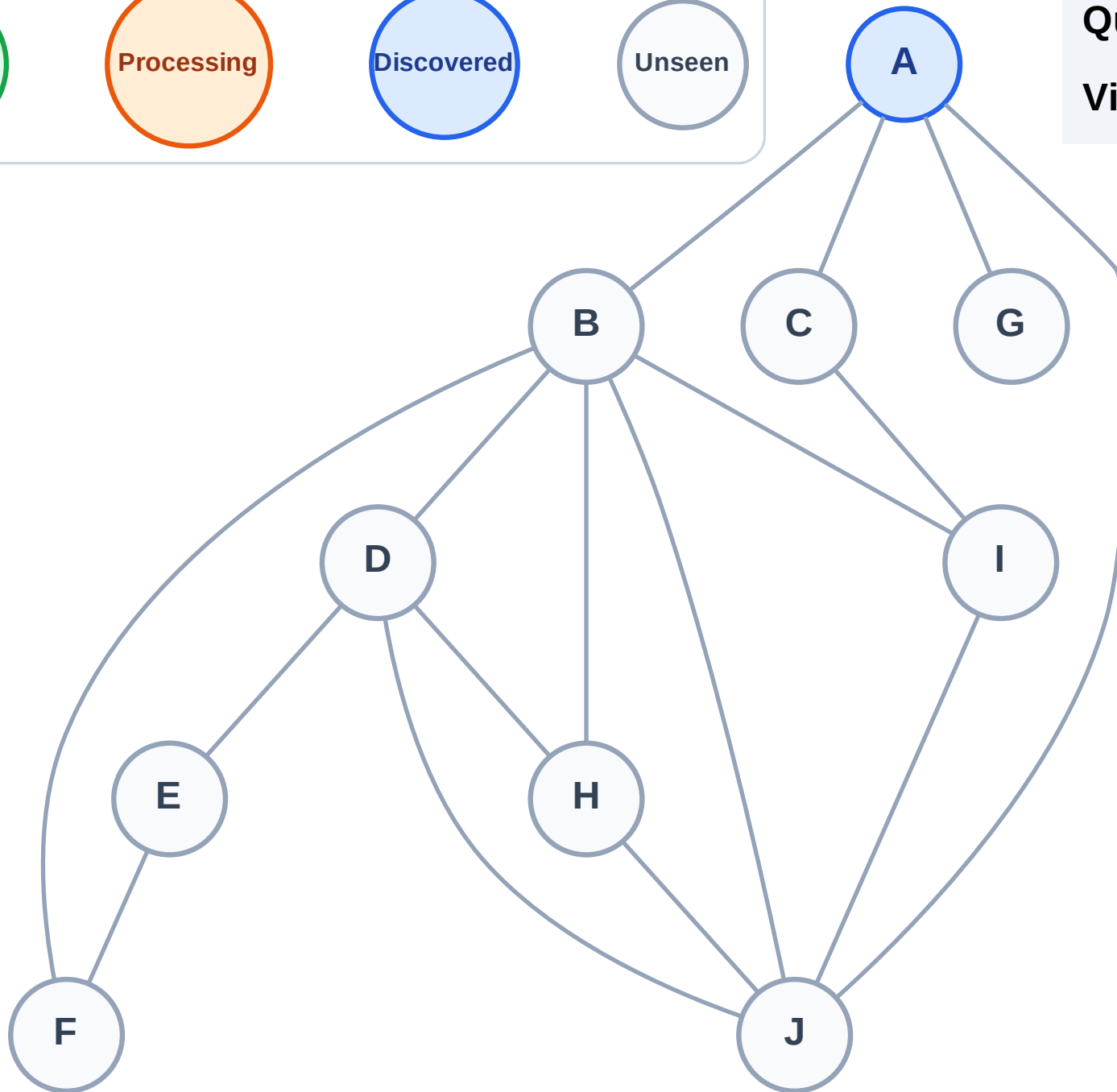
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

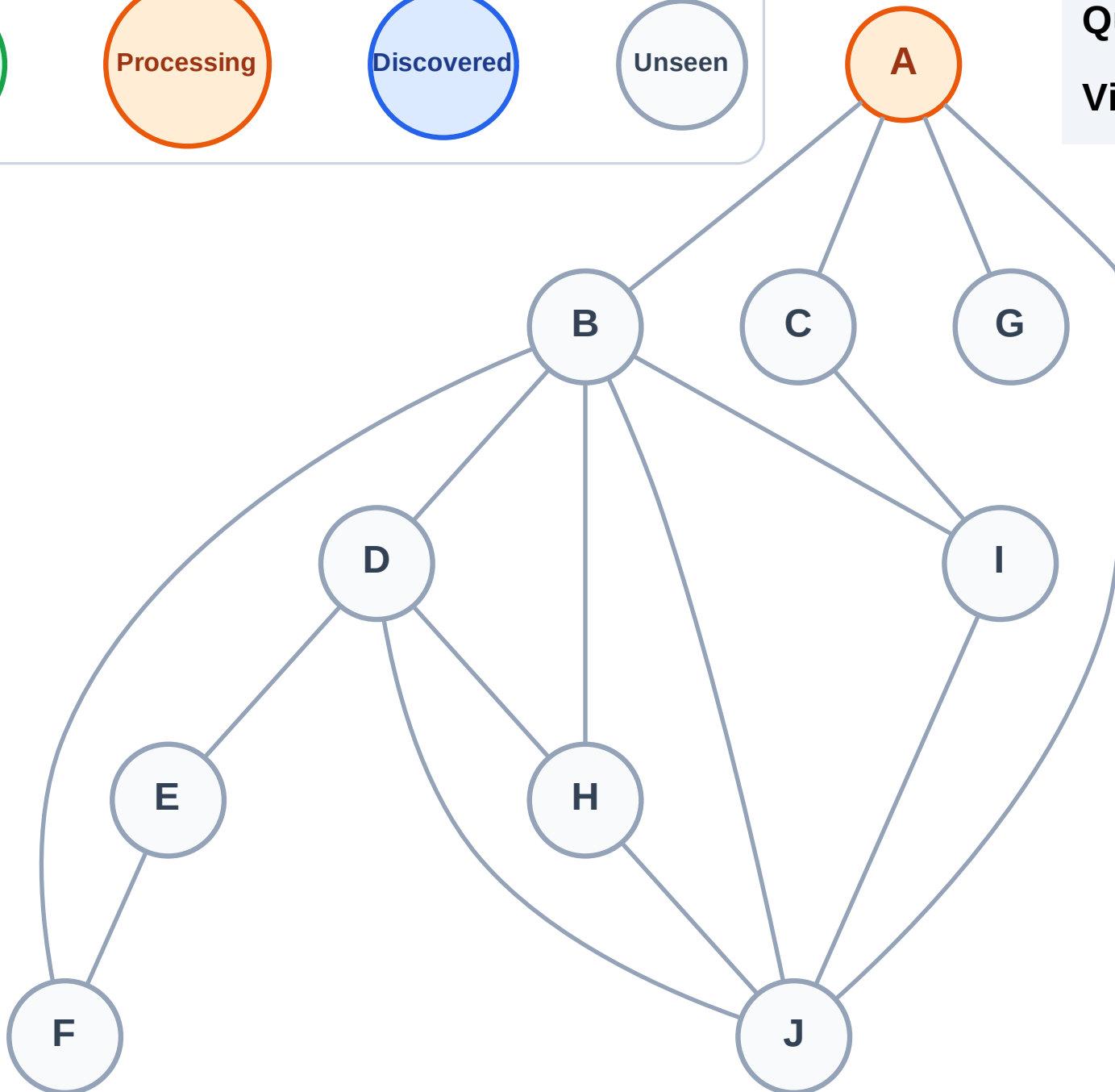
Step 2: Process node A

Legend



Queue: (empty)

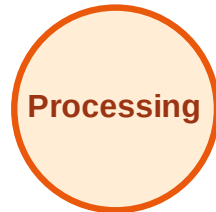
Visited: (none)



BFS Traversal

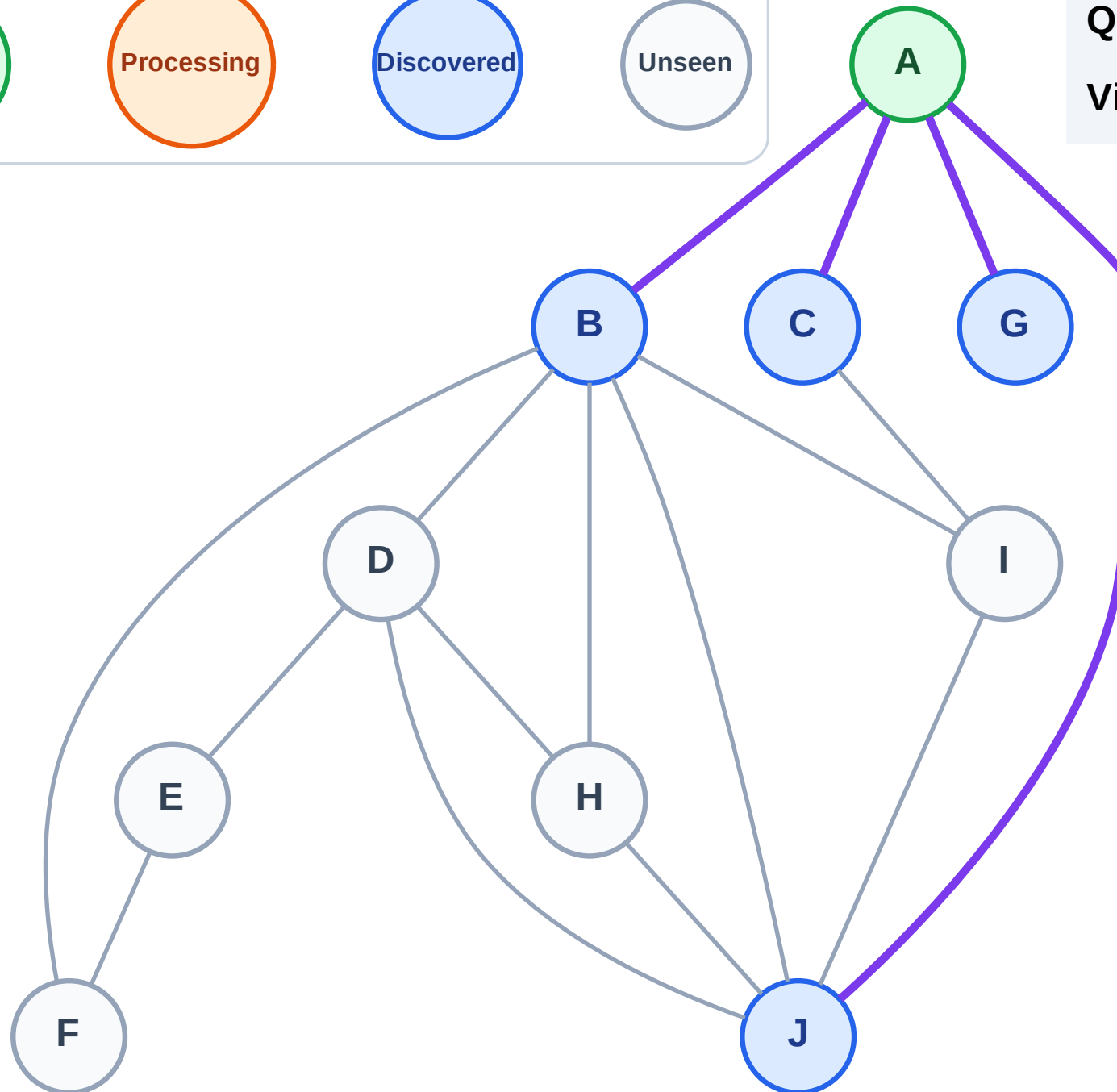
Step 3: Discover B, C, G, J from A

Legend



Queue: B → C → G → J

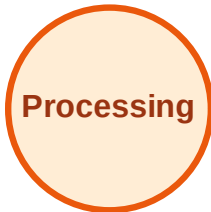
Visited: A



BFS Traversal

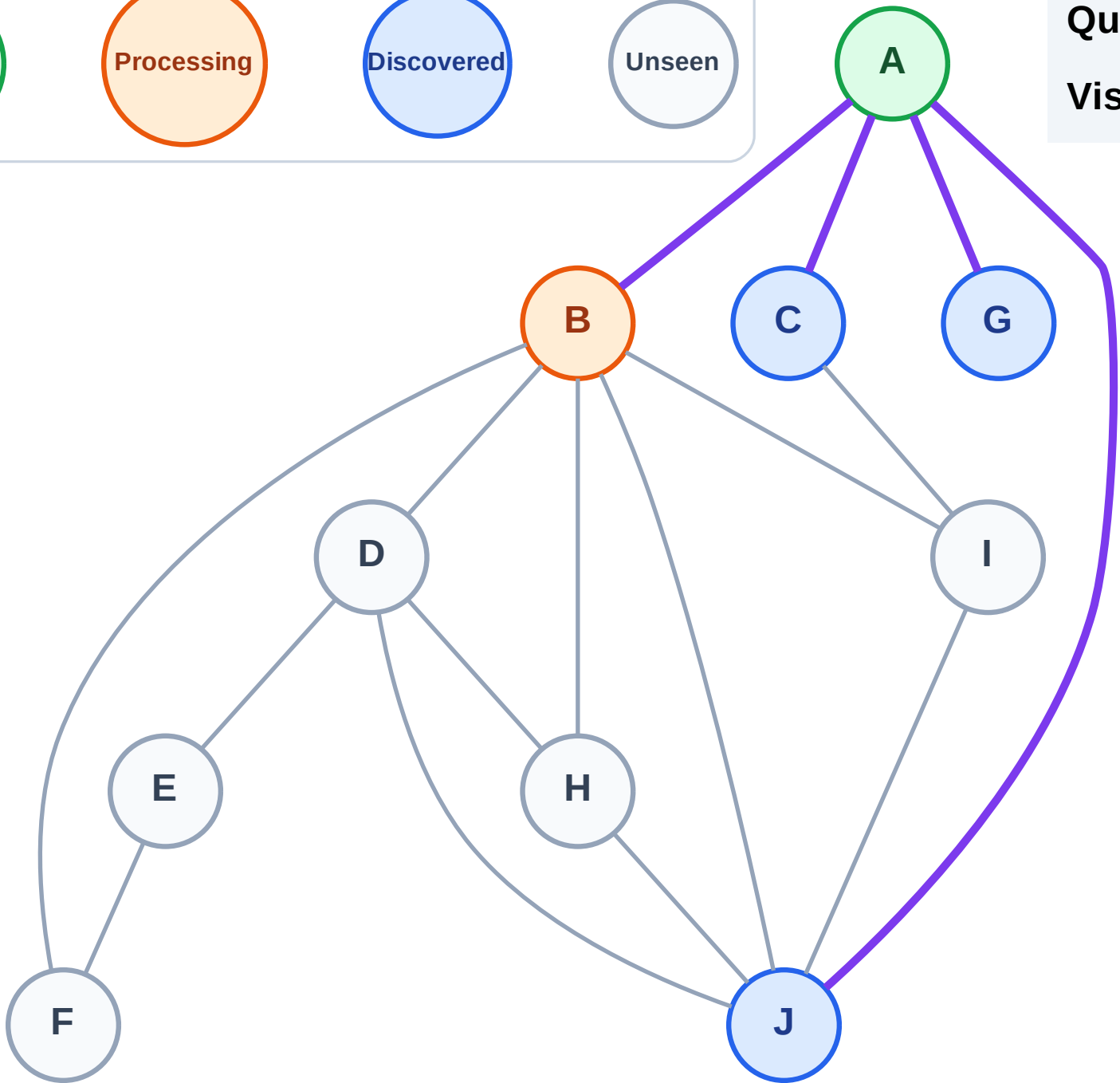
Step 4: Process node B

Legend



Queue: C → G → J

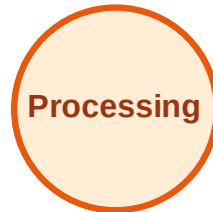
Visited: A



BFS Traversal

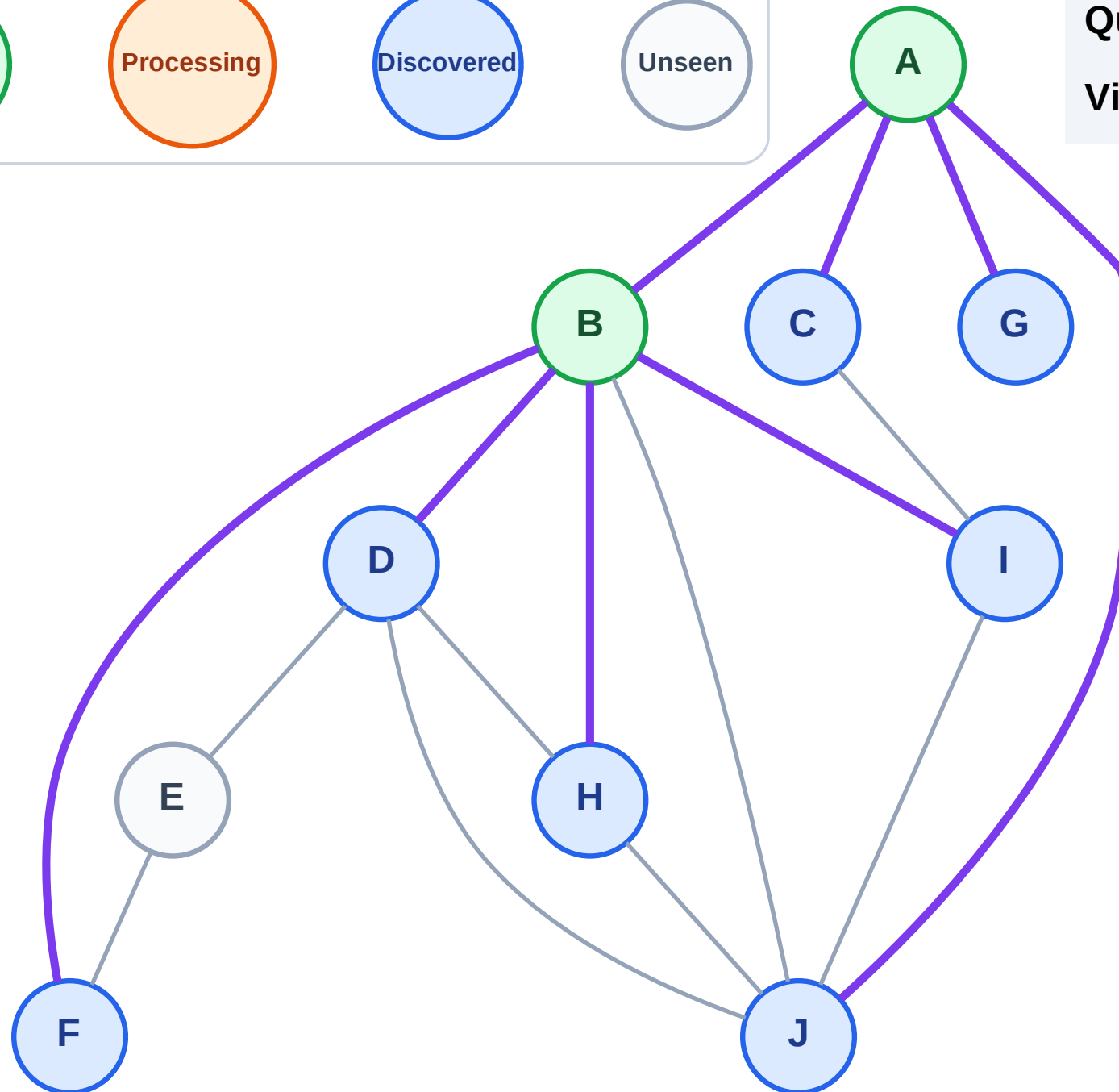
Step 5: Discover D, F, H, I from B

Legend



Queue: C → G → J → D → F → H → I

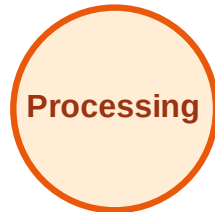
Visited: A, B



BFS Traversal

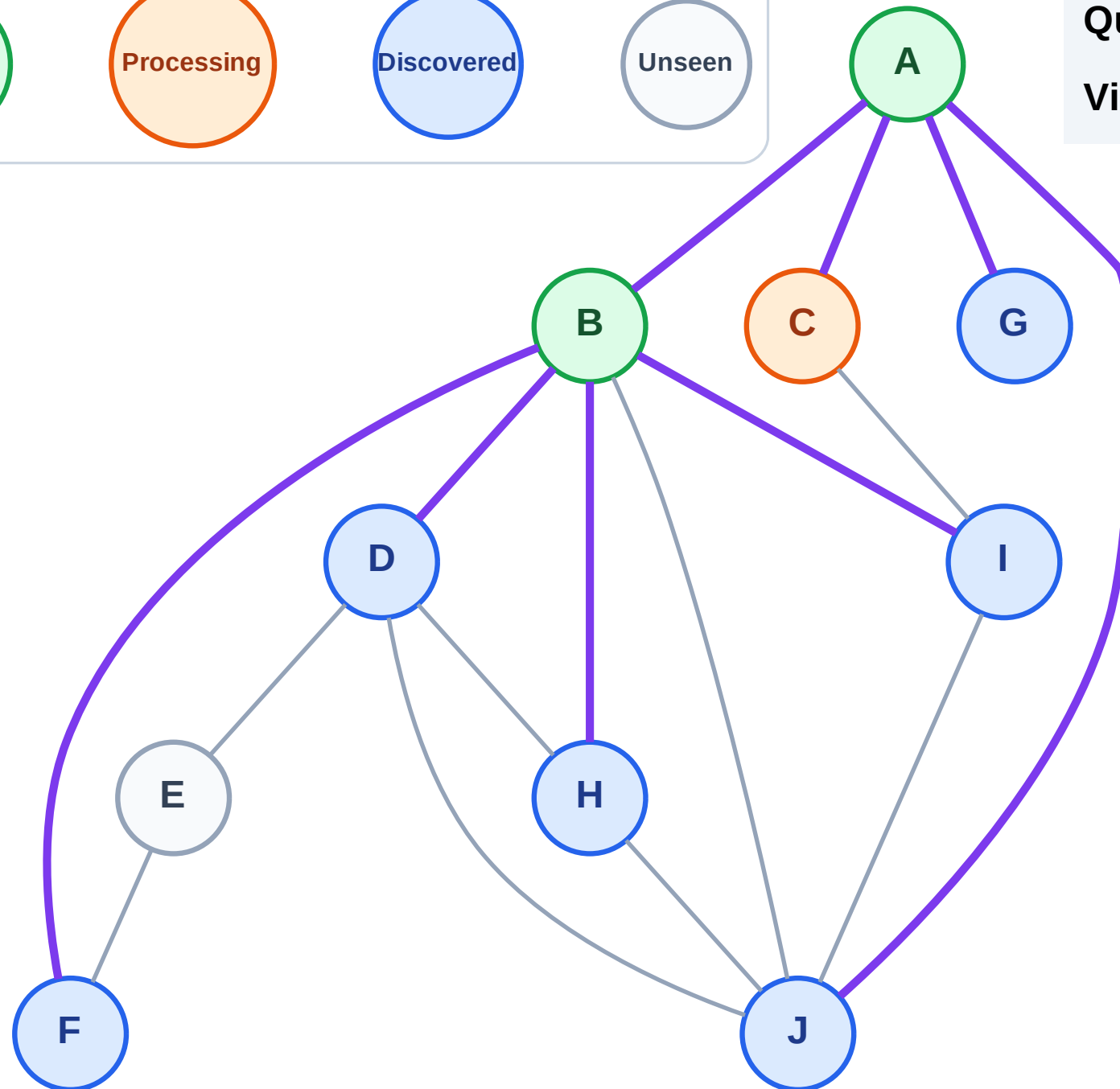
Step 6: Process node C

Legend



Queue: G → J → D → F → H → I

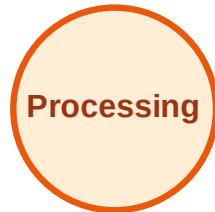
Visited: A, B



BFS Traversal

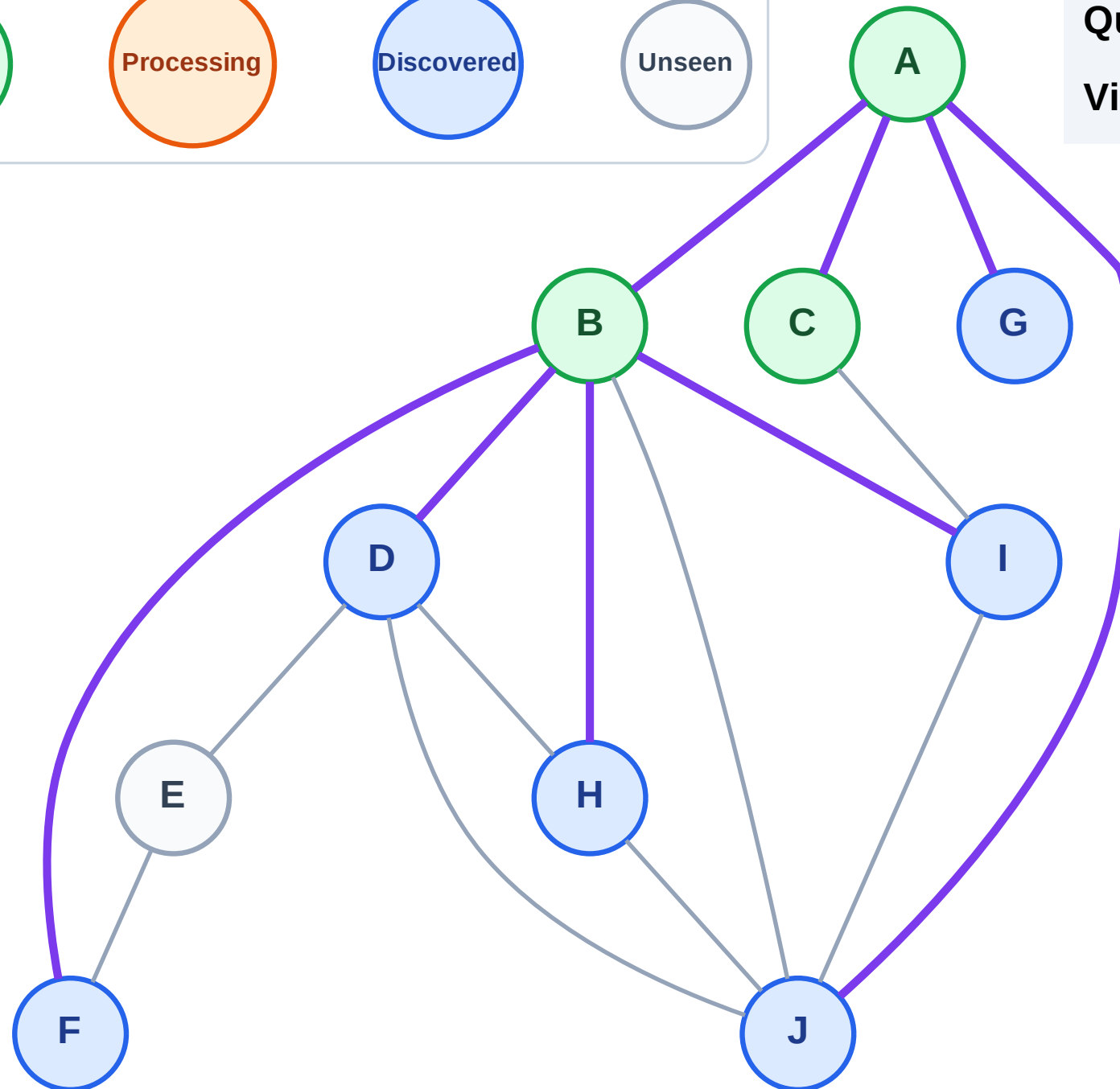
Step 7: No new neighbors from C

Legend



Queue: G → J → D → F → H → I

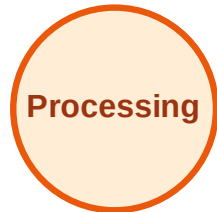
Visited: A, B, C



BFS Traversal

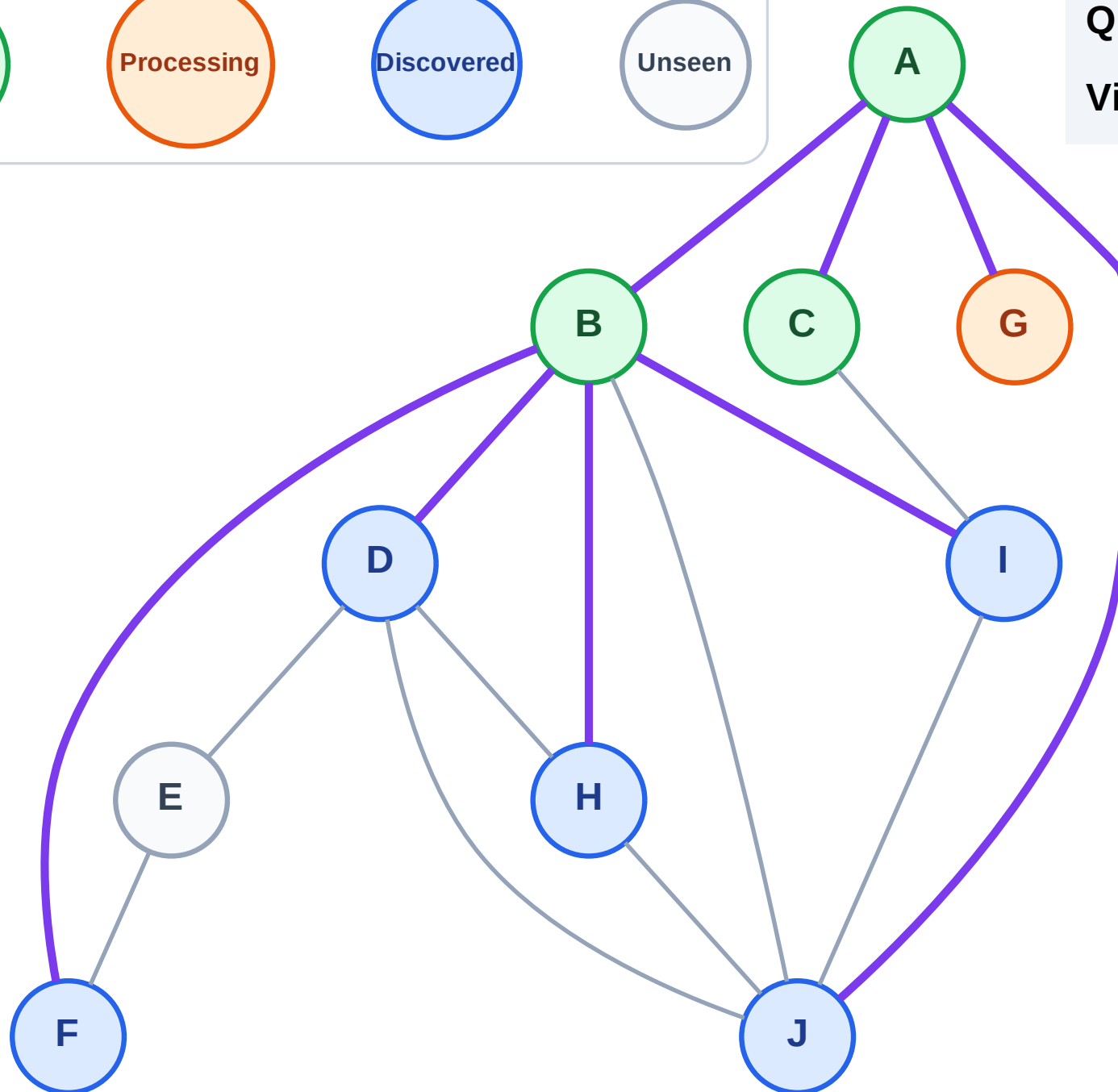
Step 8: Process node G

Legend



Queue: J → D → F → H → I

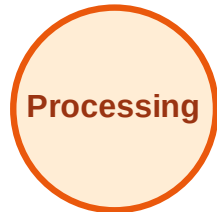
Visited: A, B, C



BFS Traversal

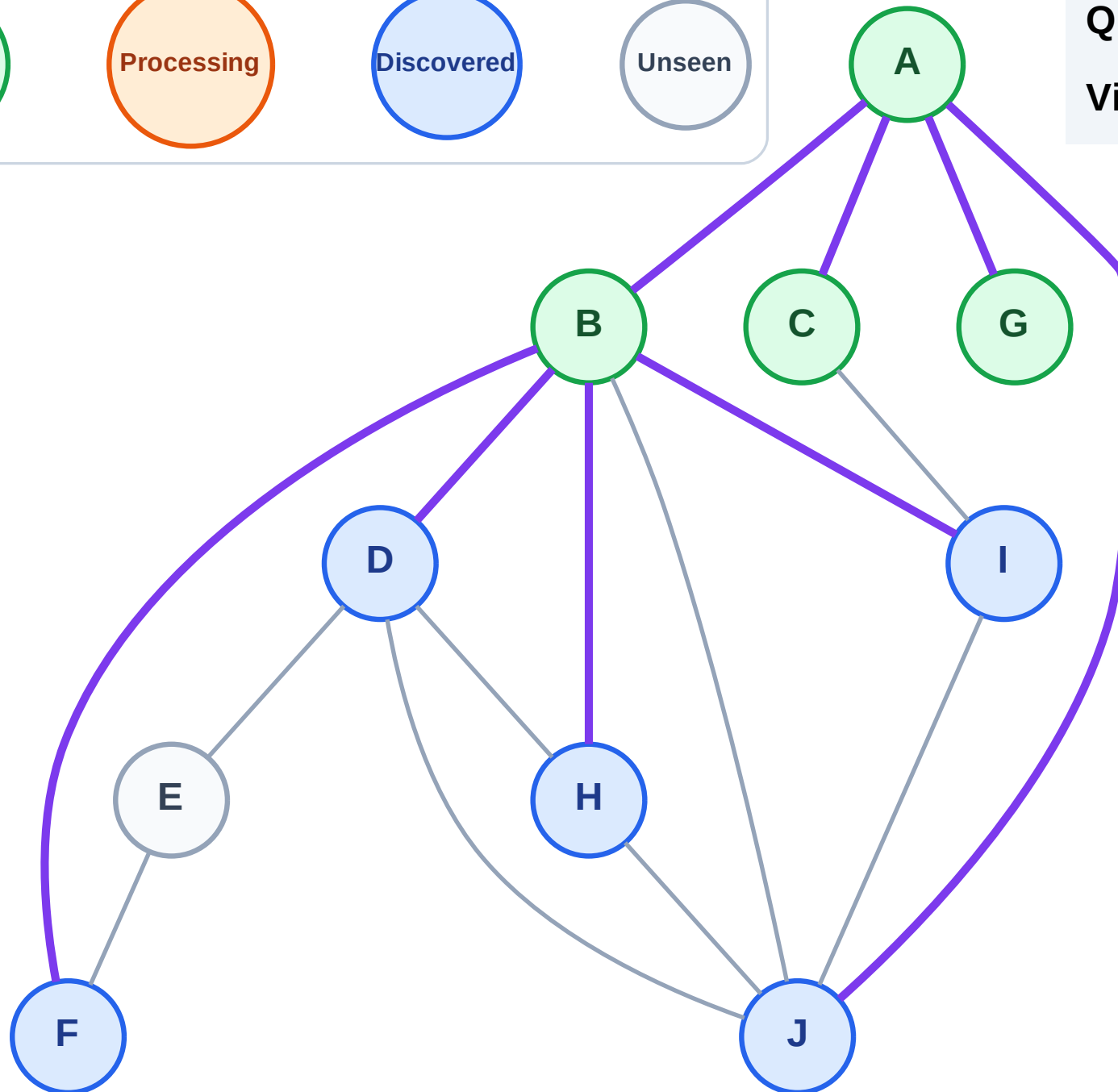
Step 9: No new neighbors from G

Legend



Queue: J → D → F → H → I

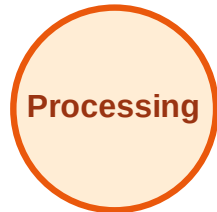
Visited: A, B, C, G



BFS Traversal

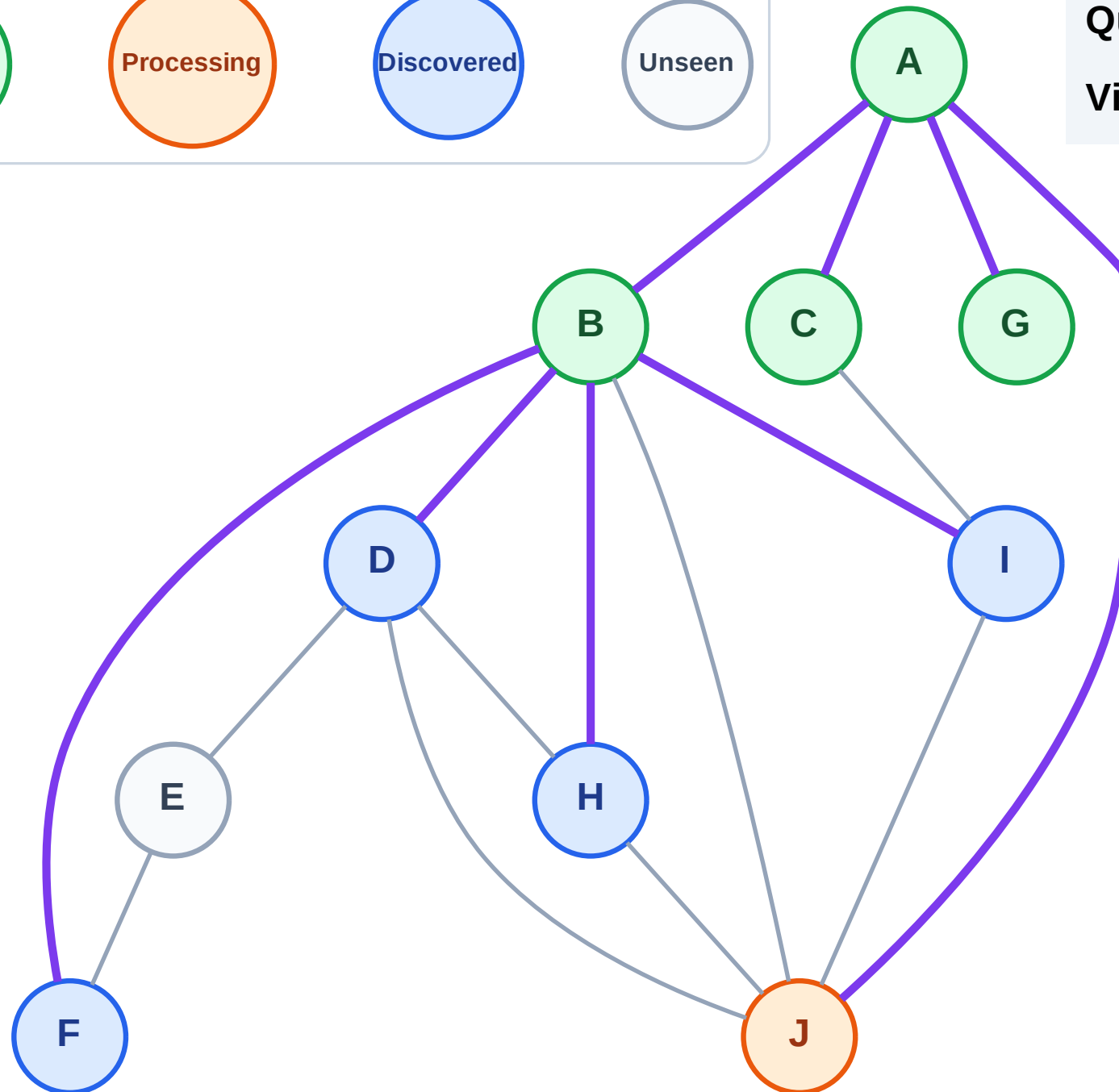
Step 10: Process node J

Legend



Queue: D → F → H → I

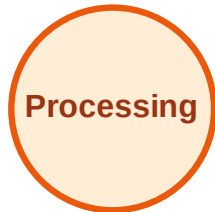
Visited: A, B, C, G



BFS Traversal

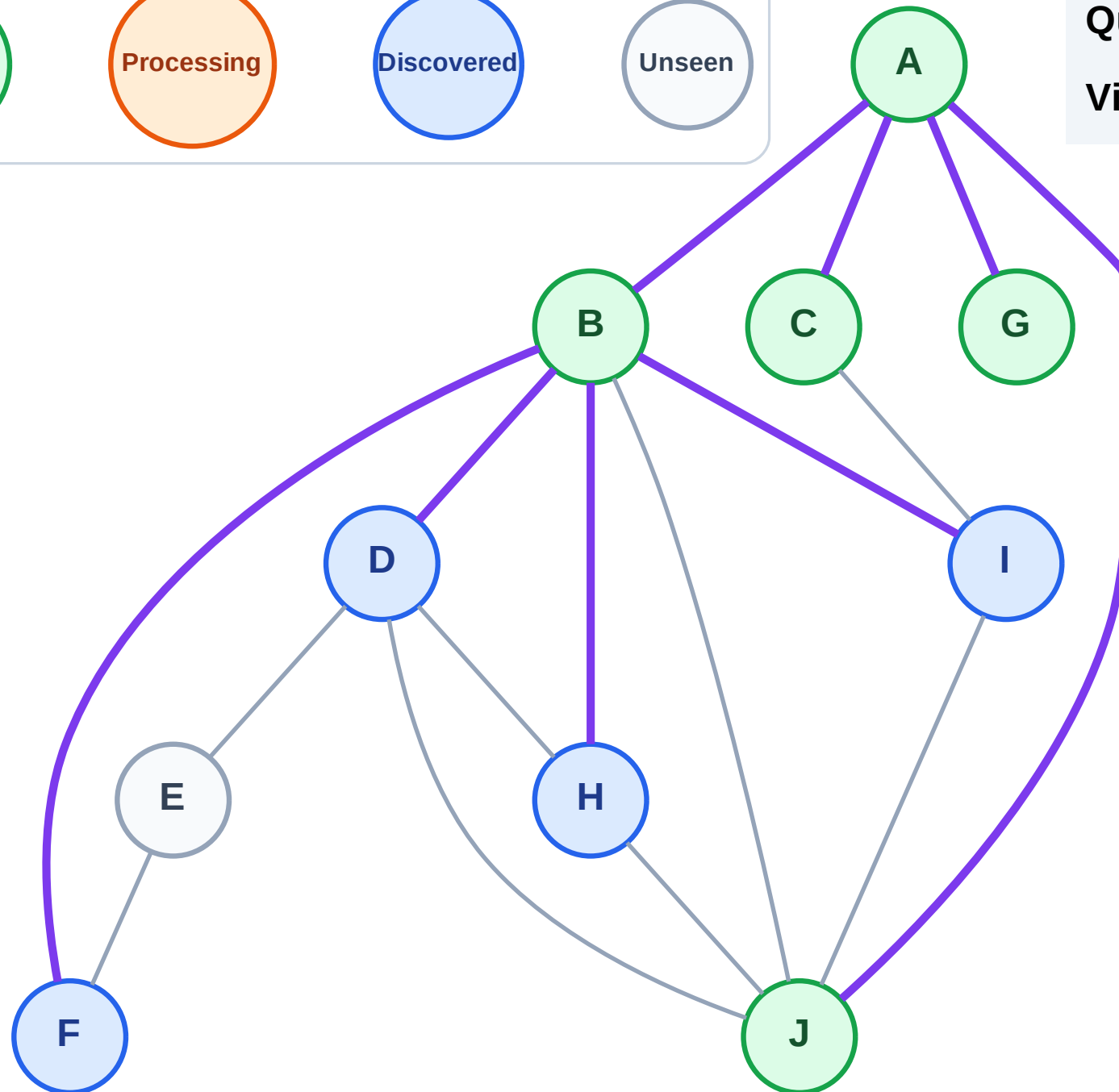
Step 11: No new neighbors from J

Legend



Queue: D → F → H → I

Visited: A, B, C, G, J



BFS Traversal

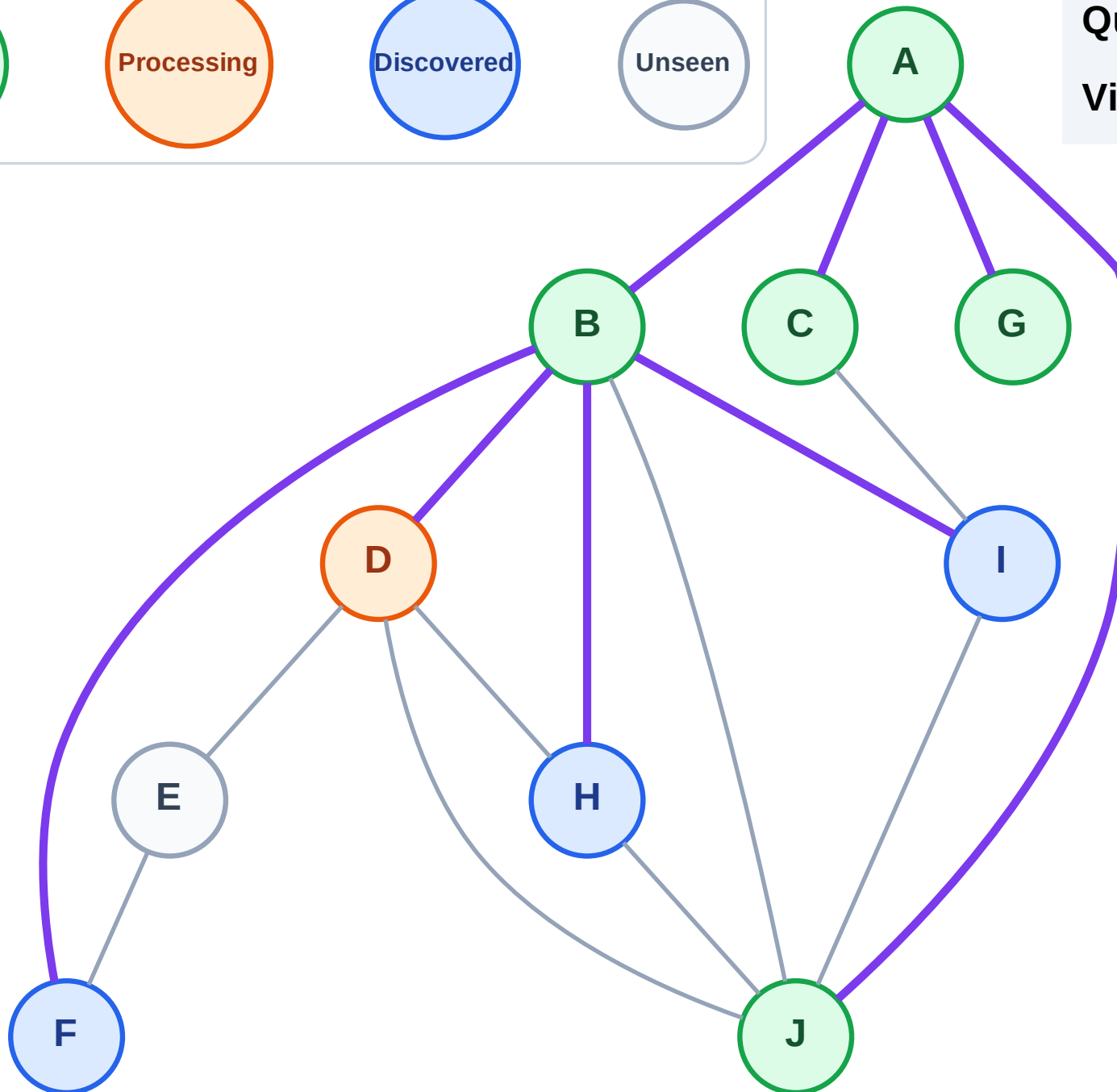
Step 12: Process node D

Legend



Queue: F → H → I

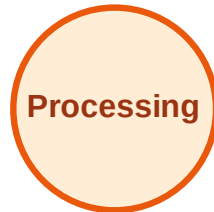
Visited: A, B, C, G, J



BFS Traversal

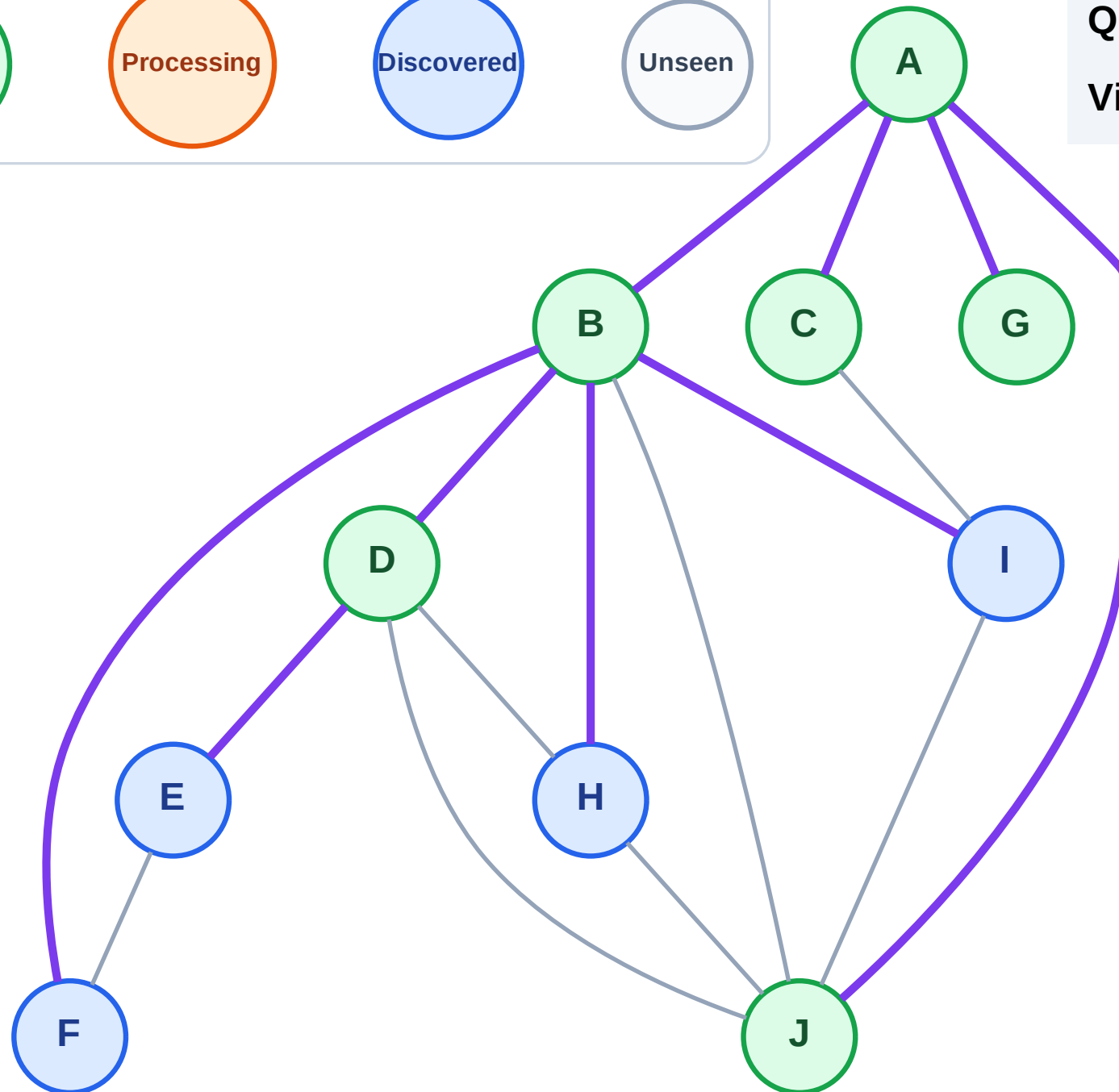
Step 13: Discover E from D

Legend



Queue: F → H → I → E

Visited: A, B, C, D, G, J



BFS Traversal

Step 14: Process node F

Legend

Complete

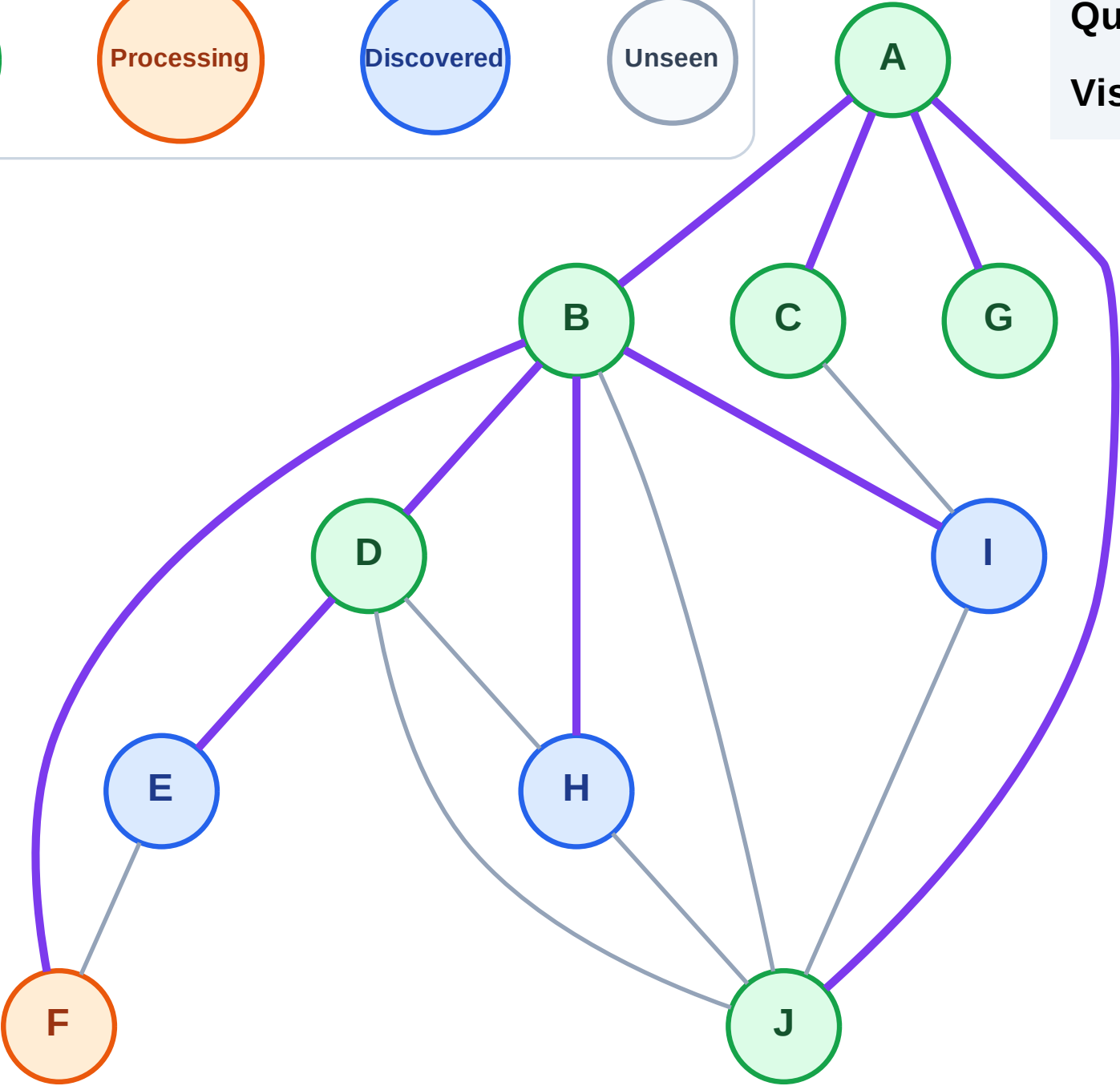
Processing

Discovered

Unseen

Queue: H → I → E

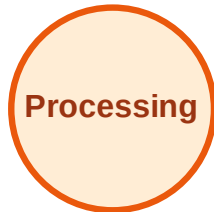
Visited: A, B, C, D, G, J



BFS Traversal

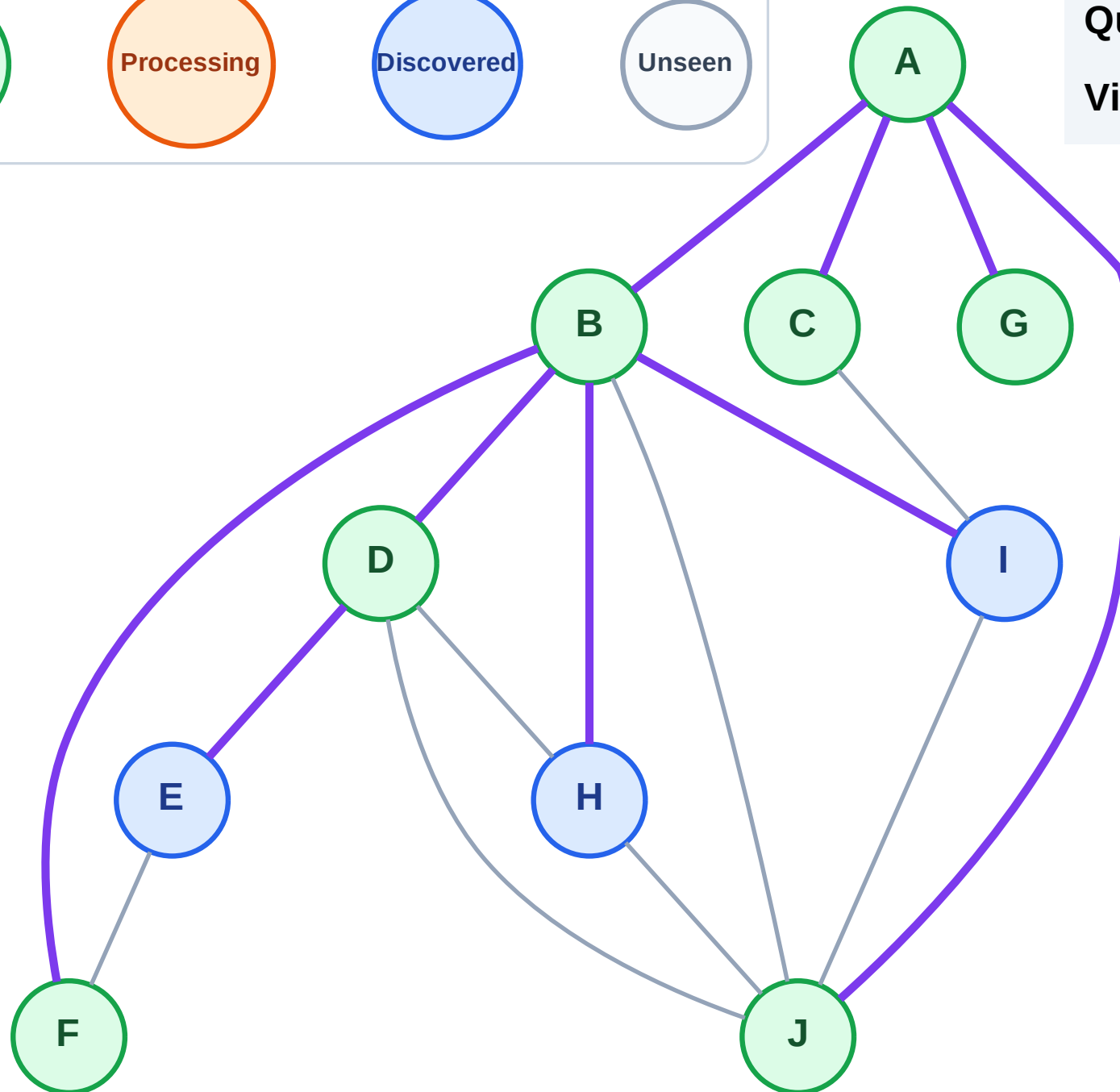
Step 15: No new neighbors from F

Legend



Queue: H → I → E

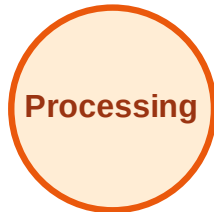
Visited: A, B, C, D, F, G, J



BFS Traversal

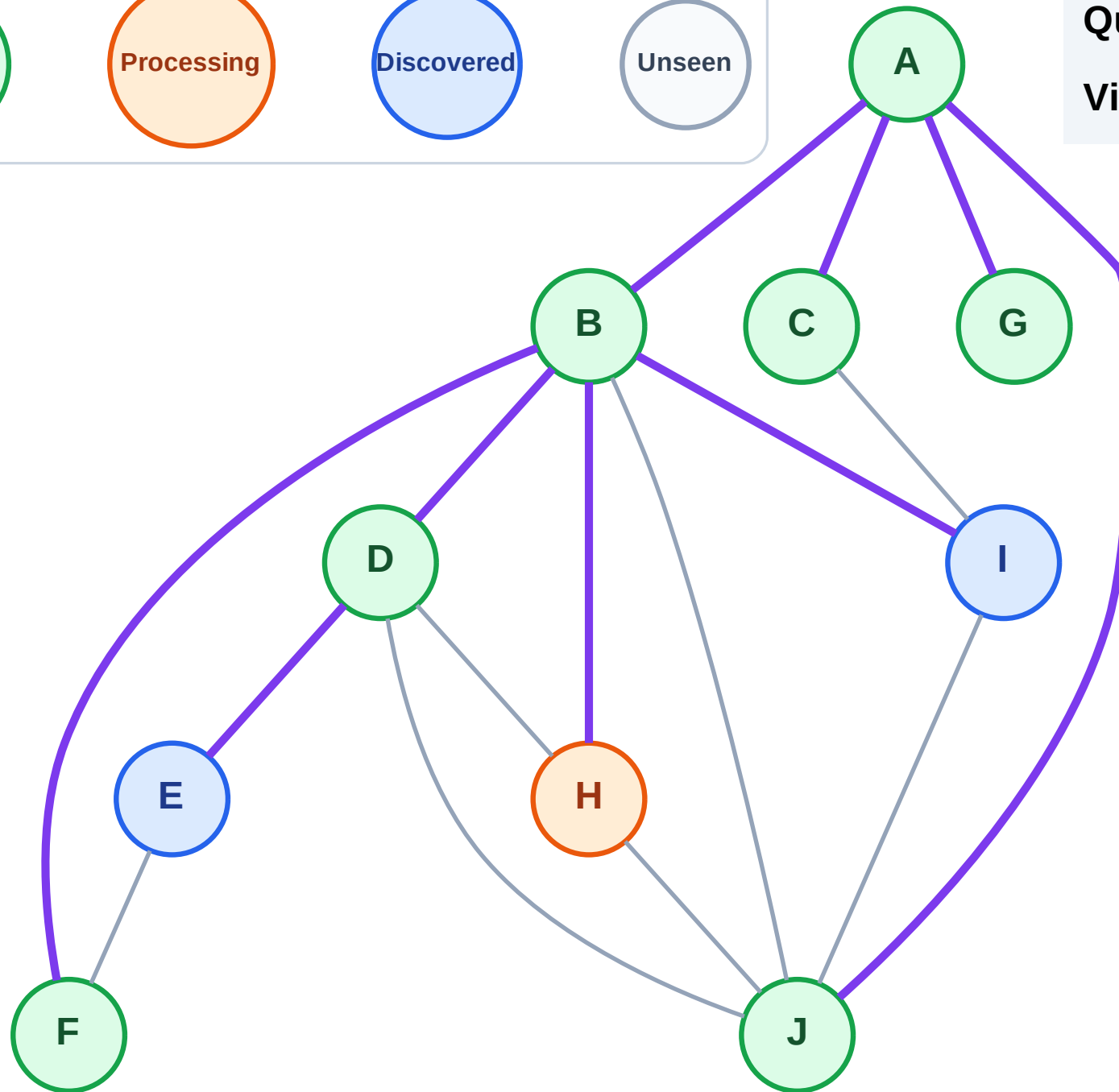
Step 16: Process node H

Legend



Queue: I → E

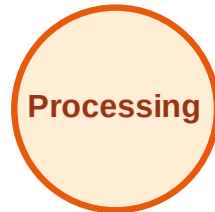
Visited: A, B, C, D, F, G, J



BFS Traversal

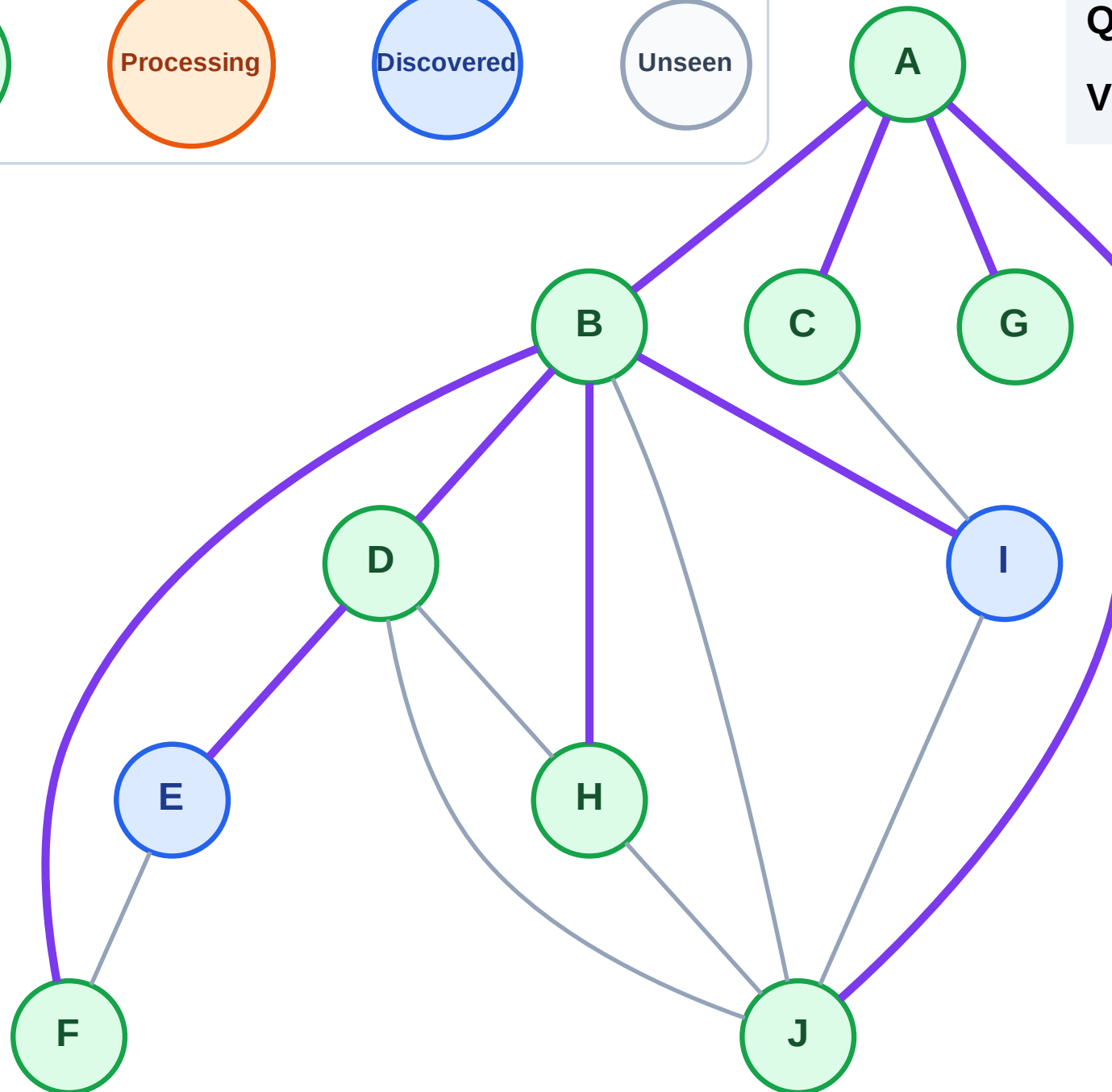
Step 17: No new neighbors from H

Legend



Queue: I → E

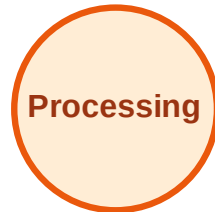
Visited: A, B, C, D, F, G, H, J



BFS Traversal

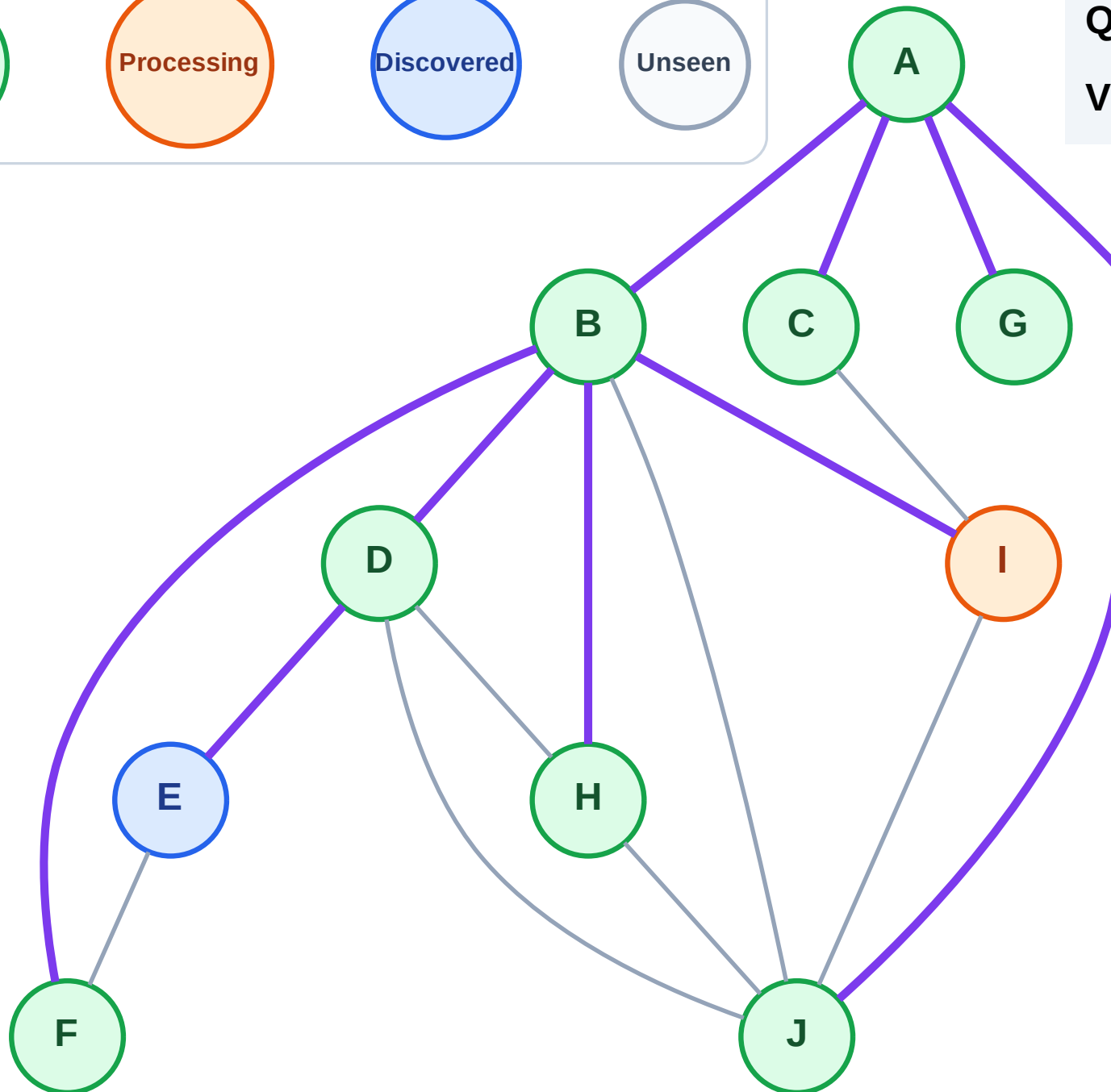
Step 18: Process node I

Legend



Queue: E

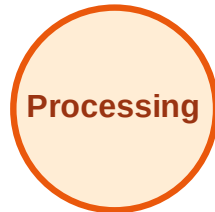
Visited: A, B, C, D, F, G, H, J



BFS Traversal

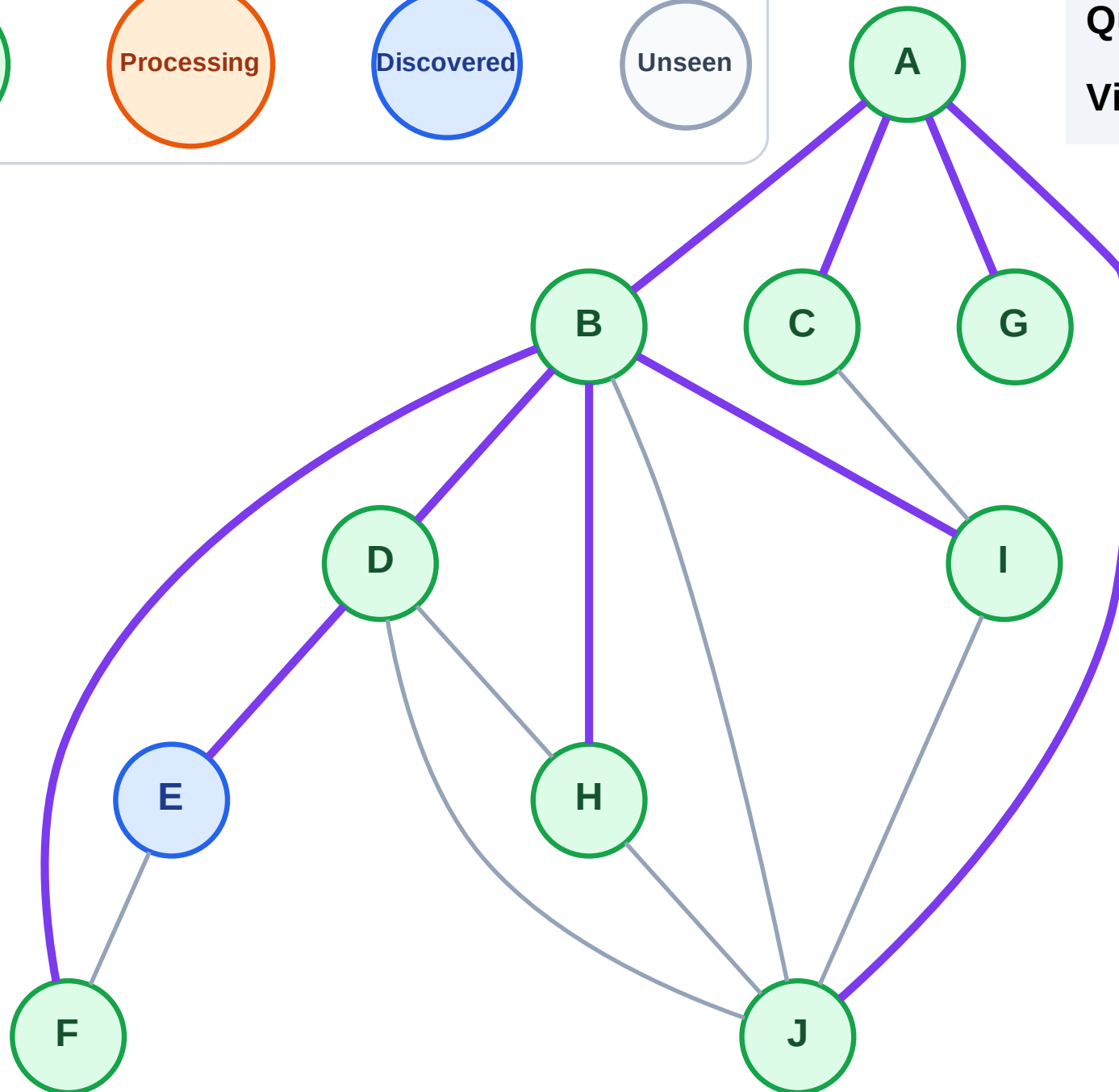
Step 19: No new neighbors from I

Legend



Queue: E

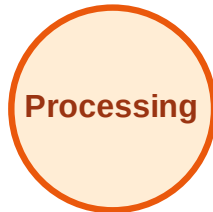
Visited: A, B, C, D, F, G, H, I, J



BFS Traversal

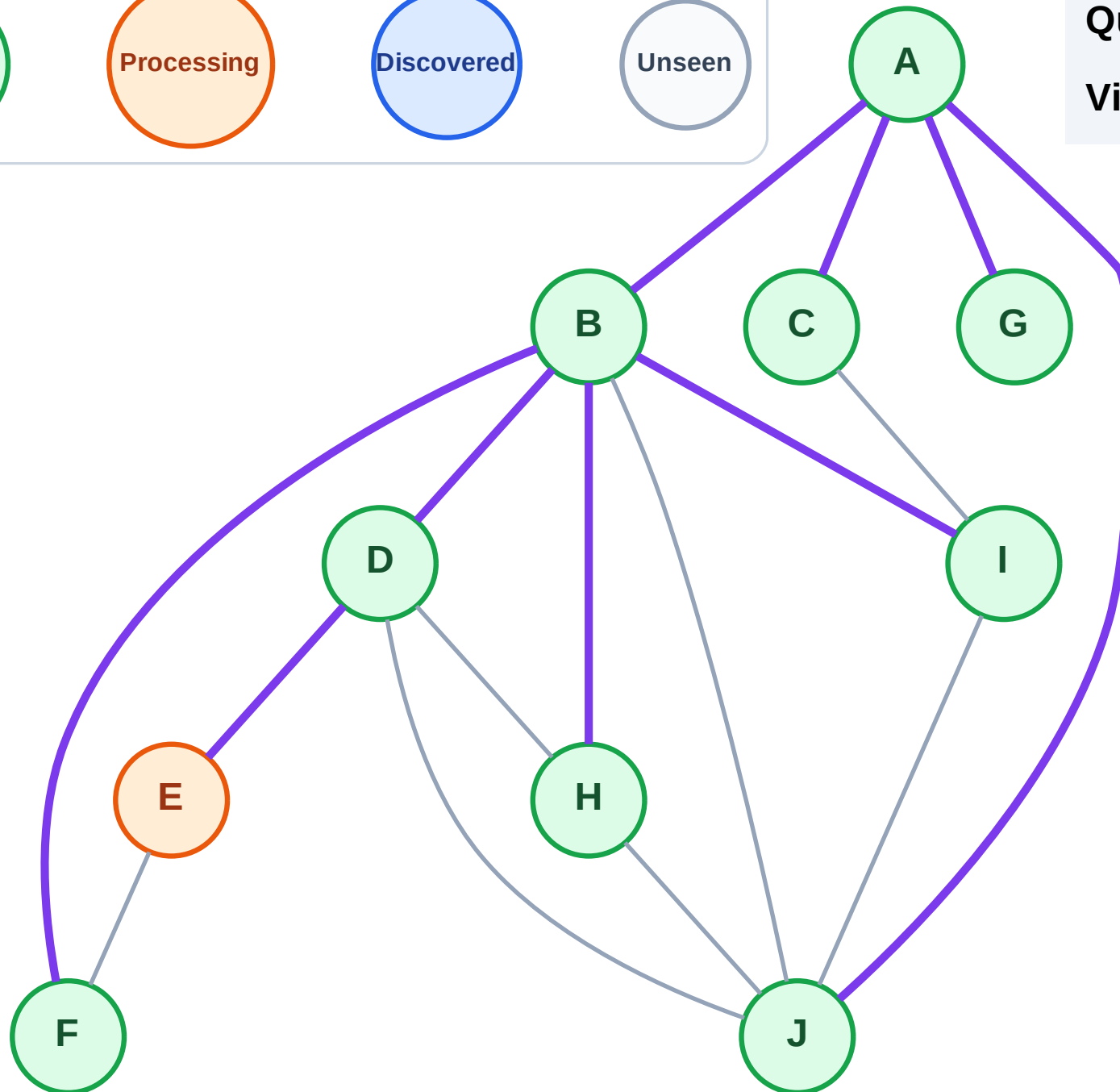
Step 20: Process node E

Legend



Queue: (empty)

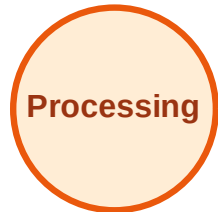
Visited: A, B, C, D, F, G, H, I, J



BFS Traversal

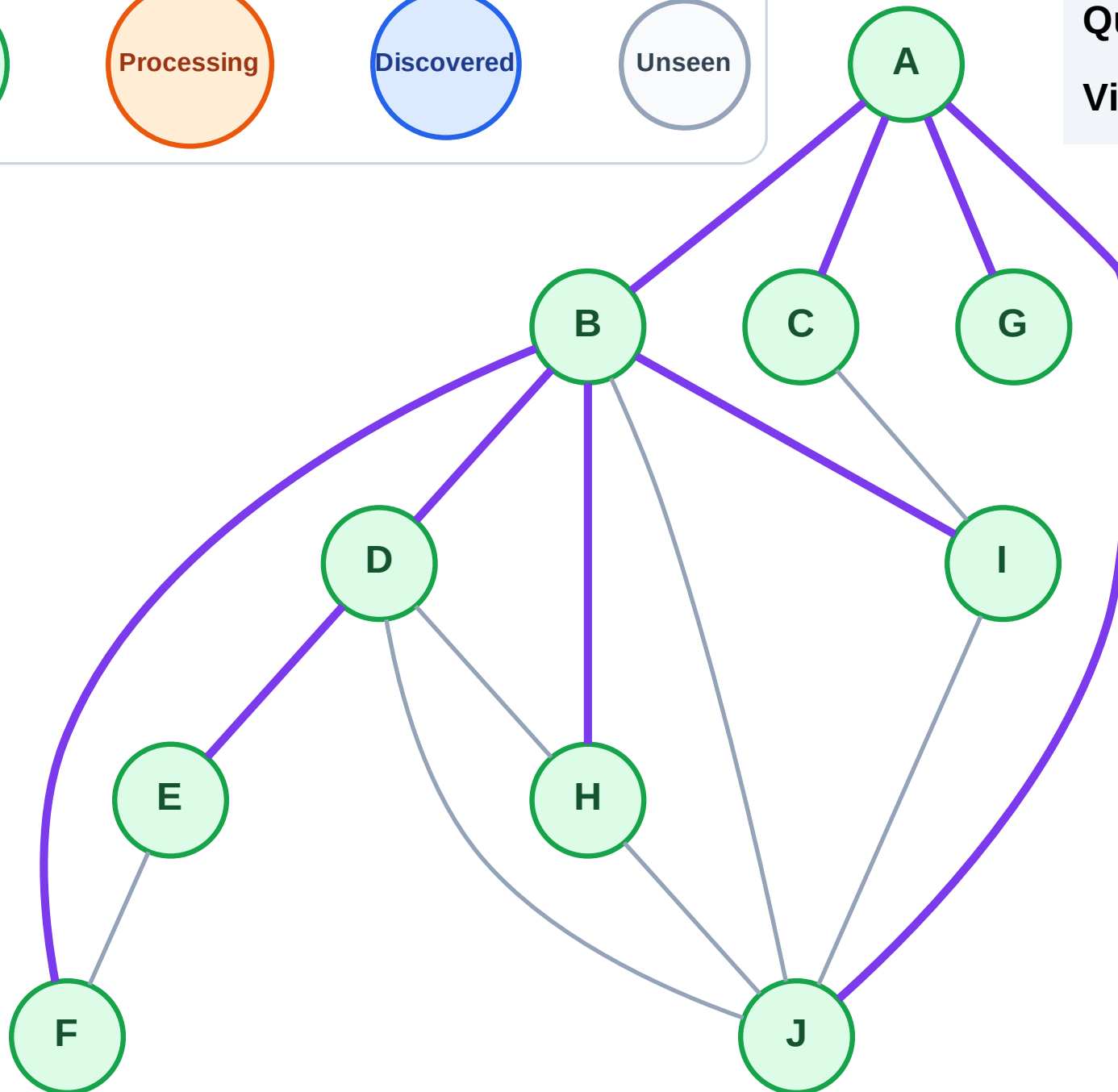
Step 21: No new neighbors from E

Legend



Queue: (empty)

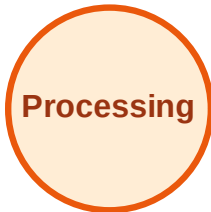
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

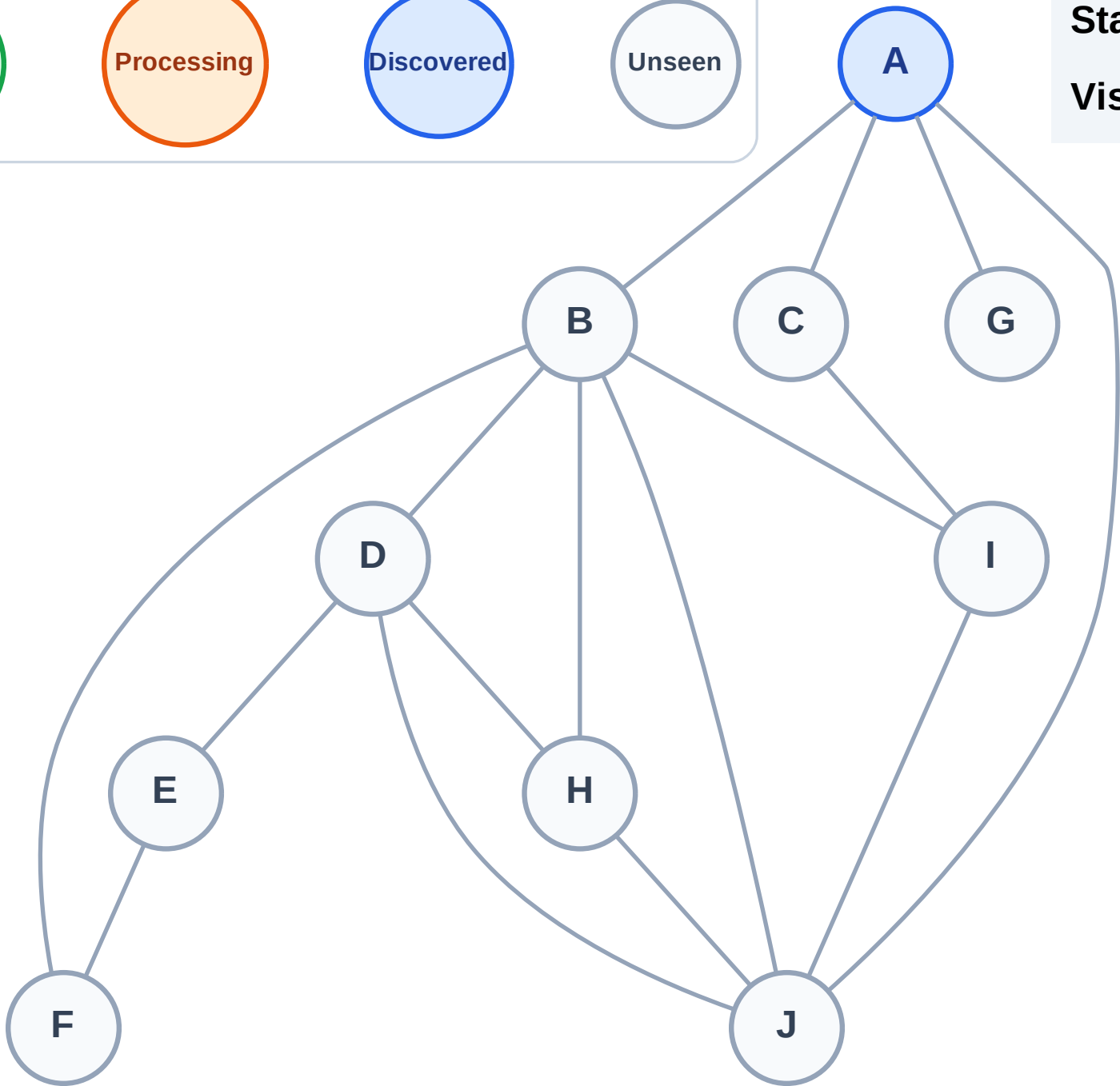
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

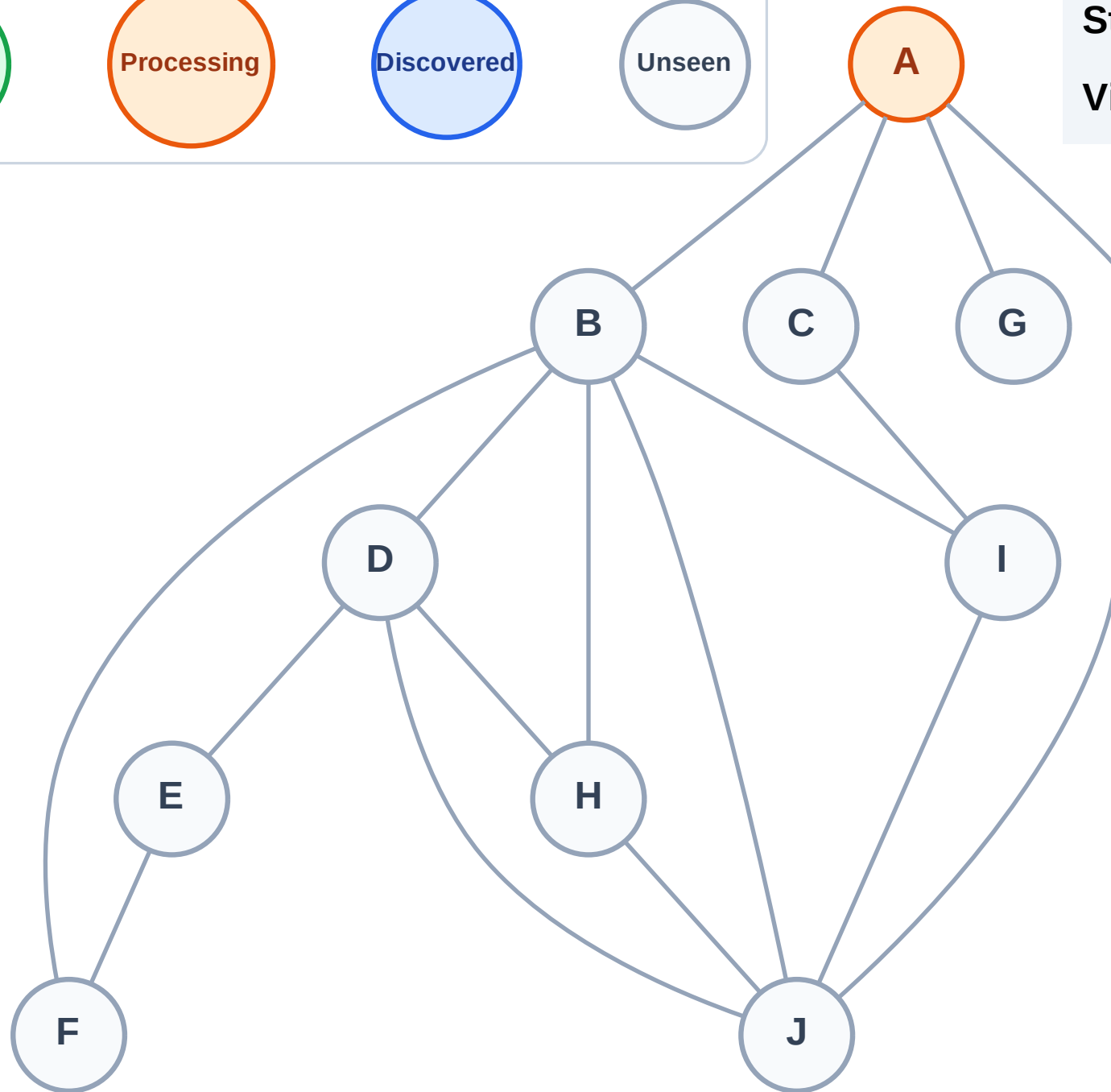
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover J, G, C, B from A

Legend

Complete

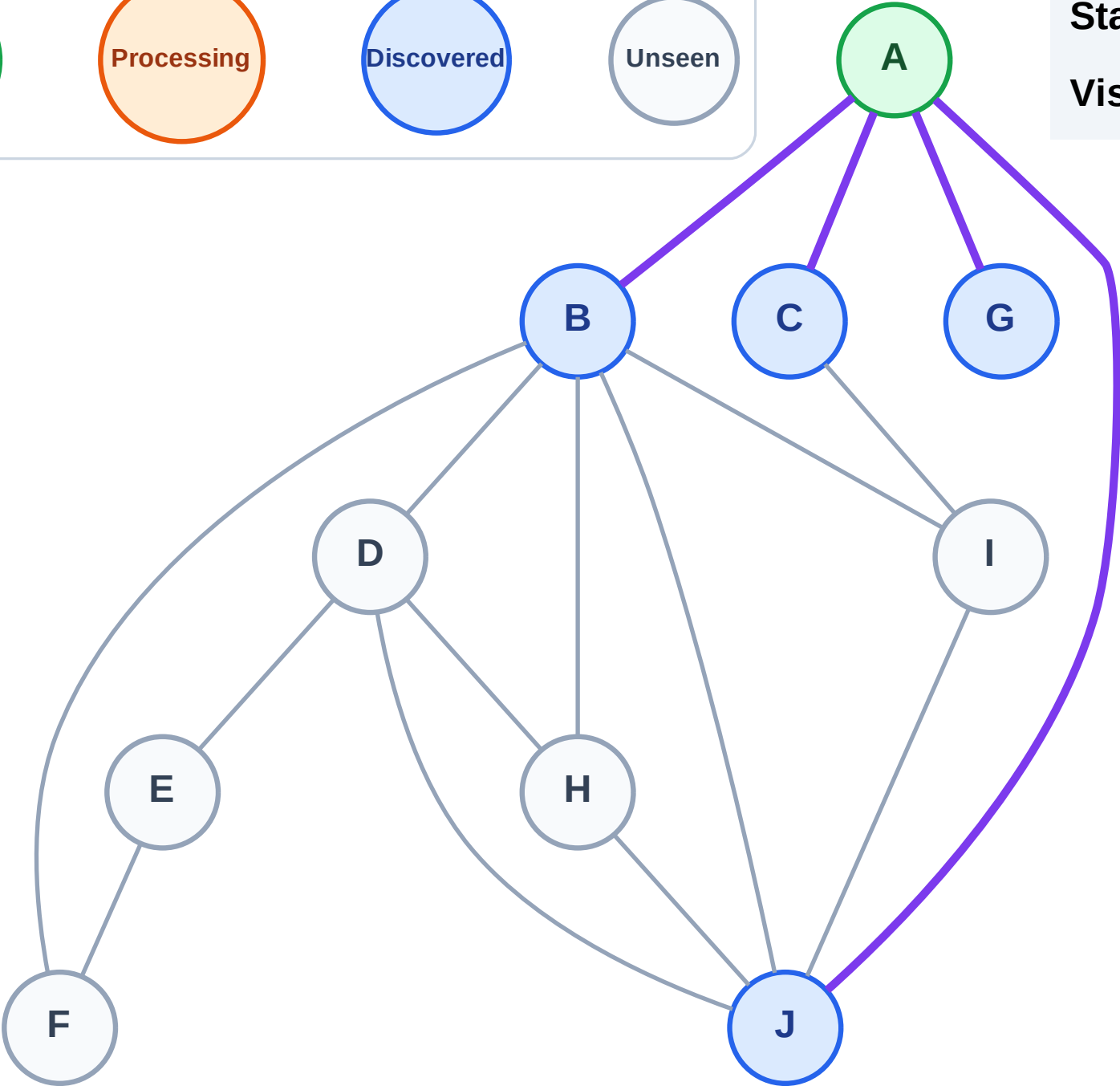
Processing

Discovered

Unseen

Stack (top at right): J | G | C | B

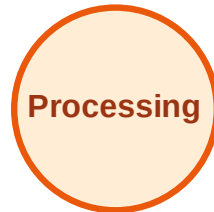
Visited: A



DFS Traversal

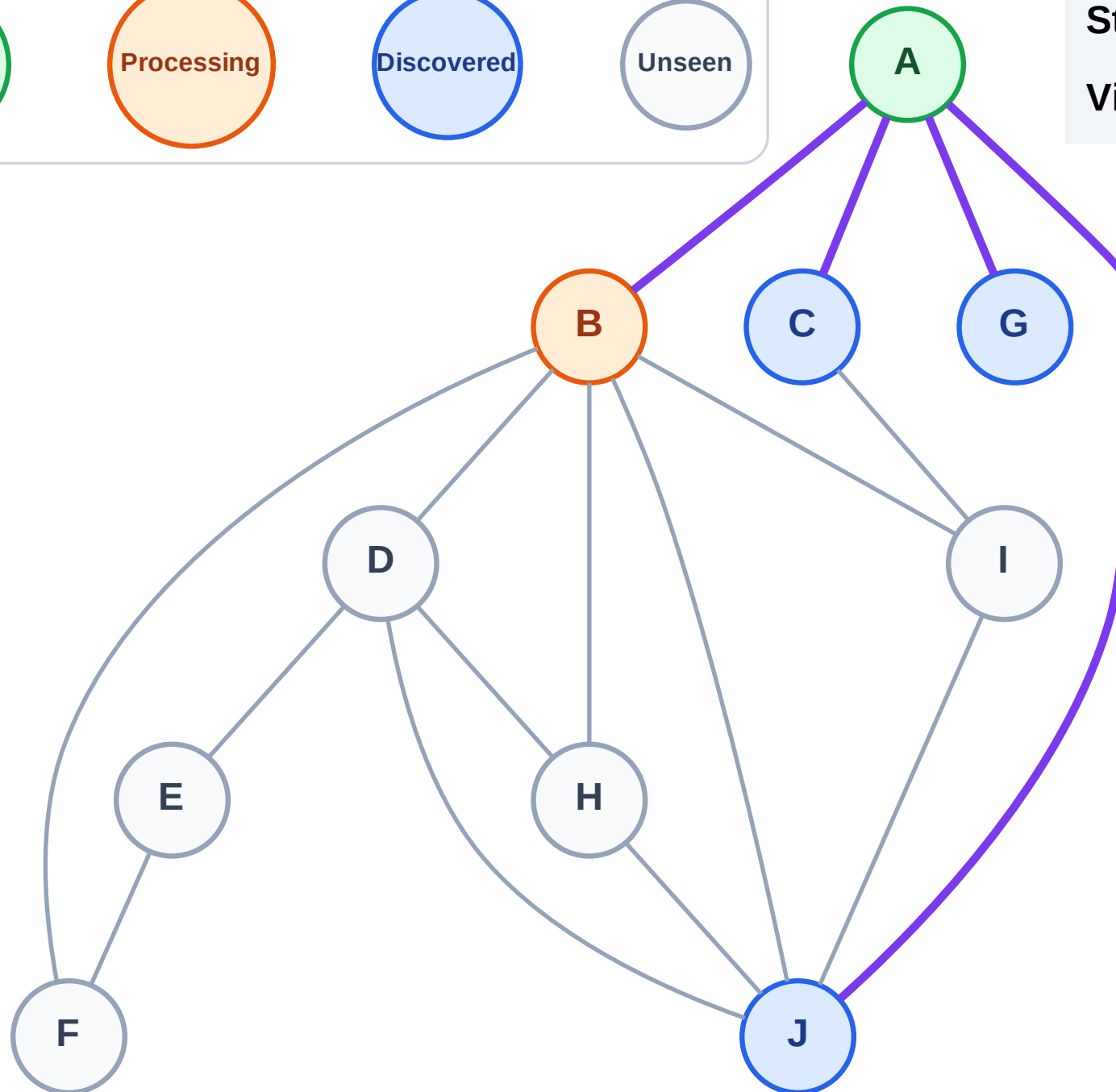
Step 4: Process node B

Legend



Stack (top at right): J | G | C

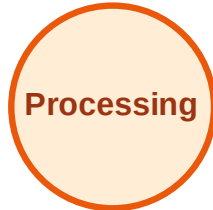
Visited: A



DFS Traversal

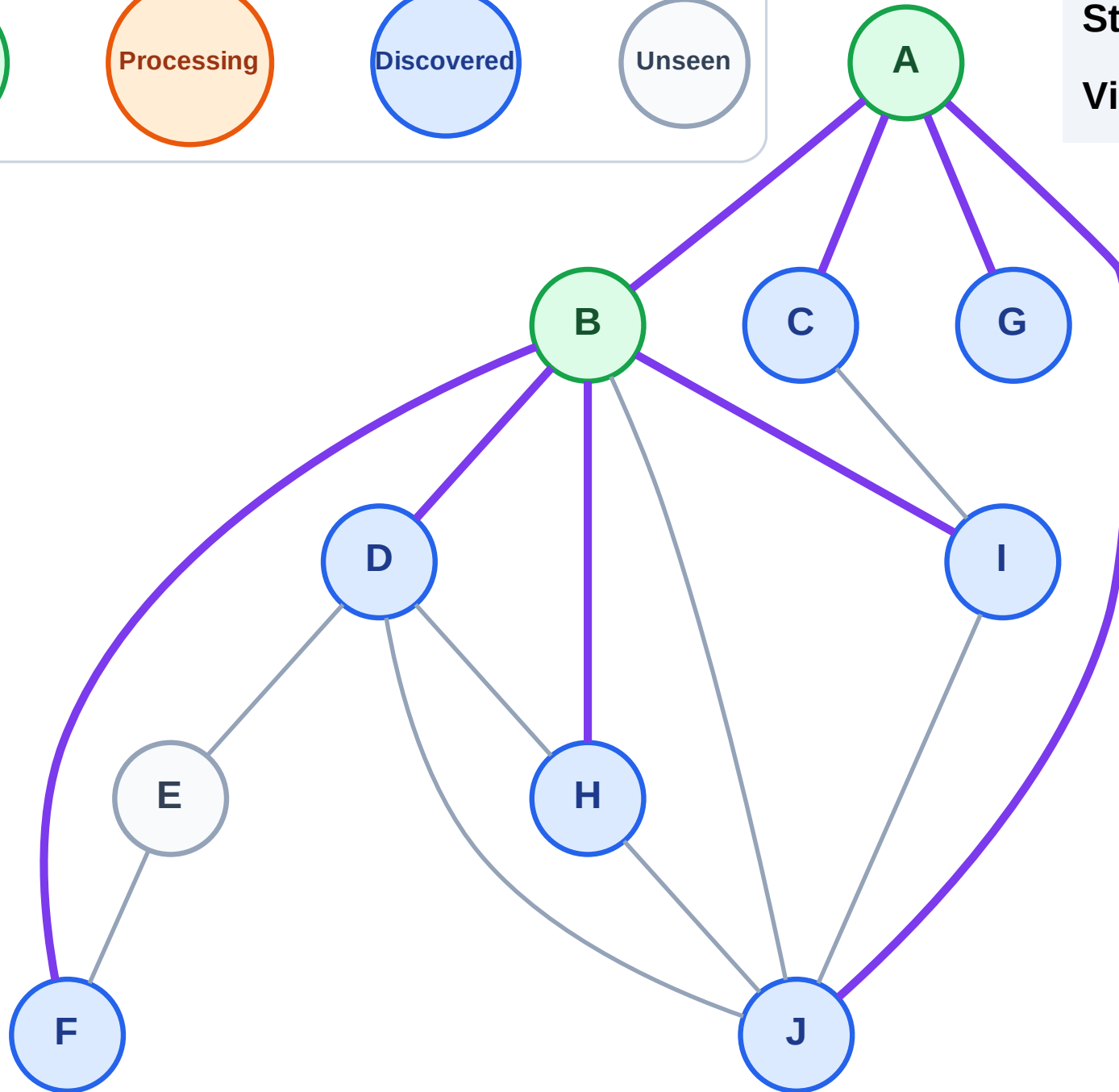
Step 5: Discover I, H, F, D from B

Legend



Stack (top at right): J | G | C | I | H | F | D

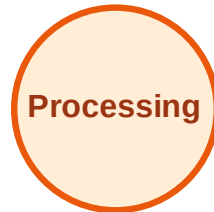
Visited: A, B



DFS Traversal

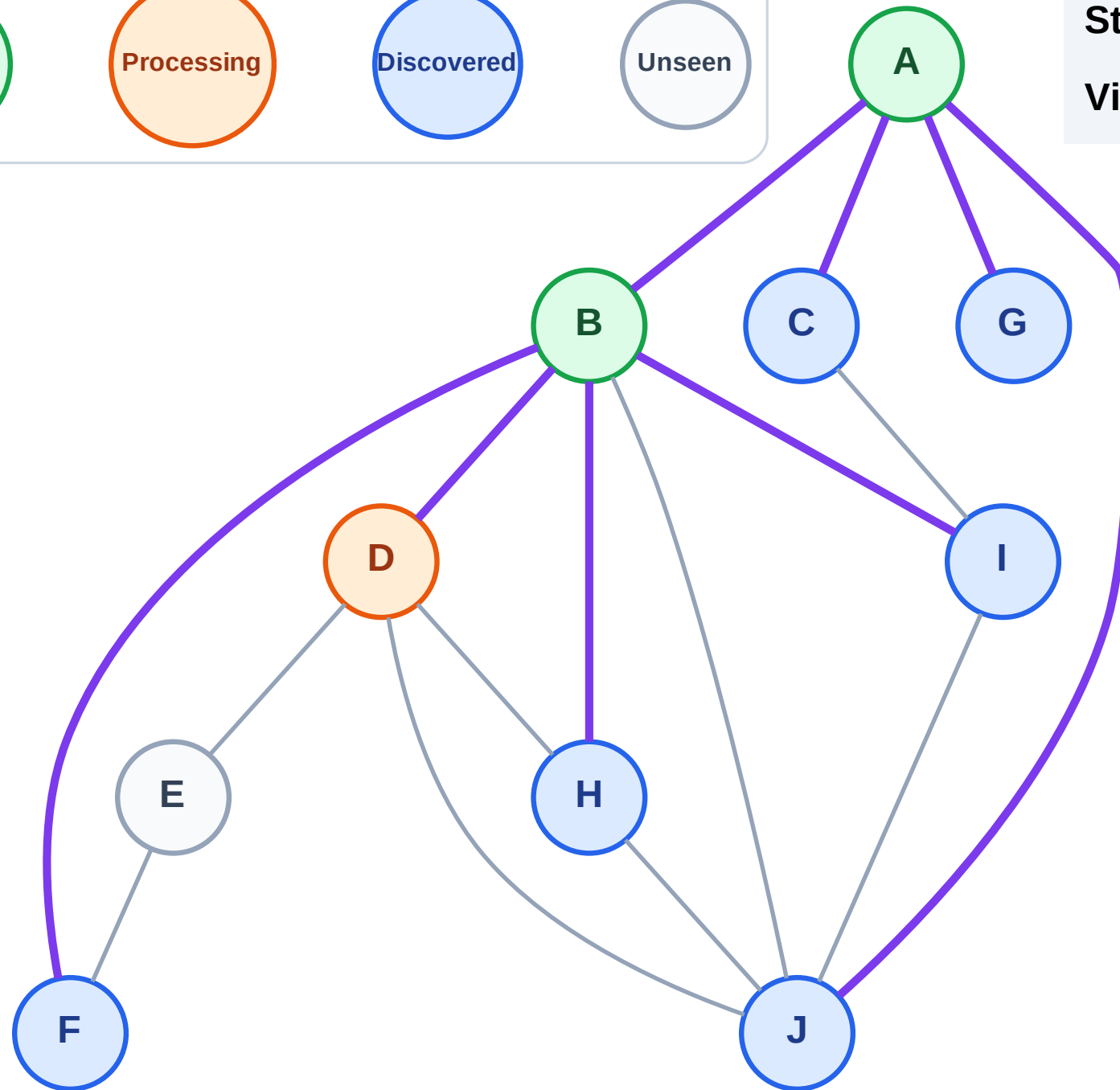
Step 6: Process node D

Legend



Stack (top at right): J | G | C | I | H | F

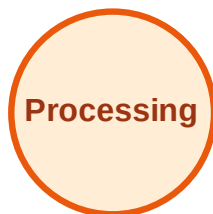
Visited: A, B



DFS Traversal

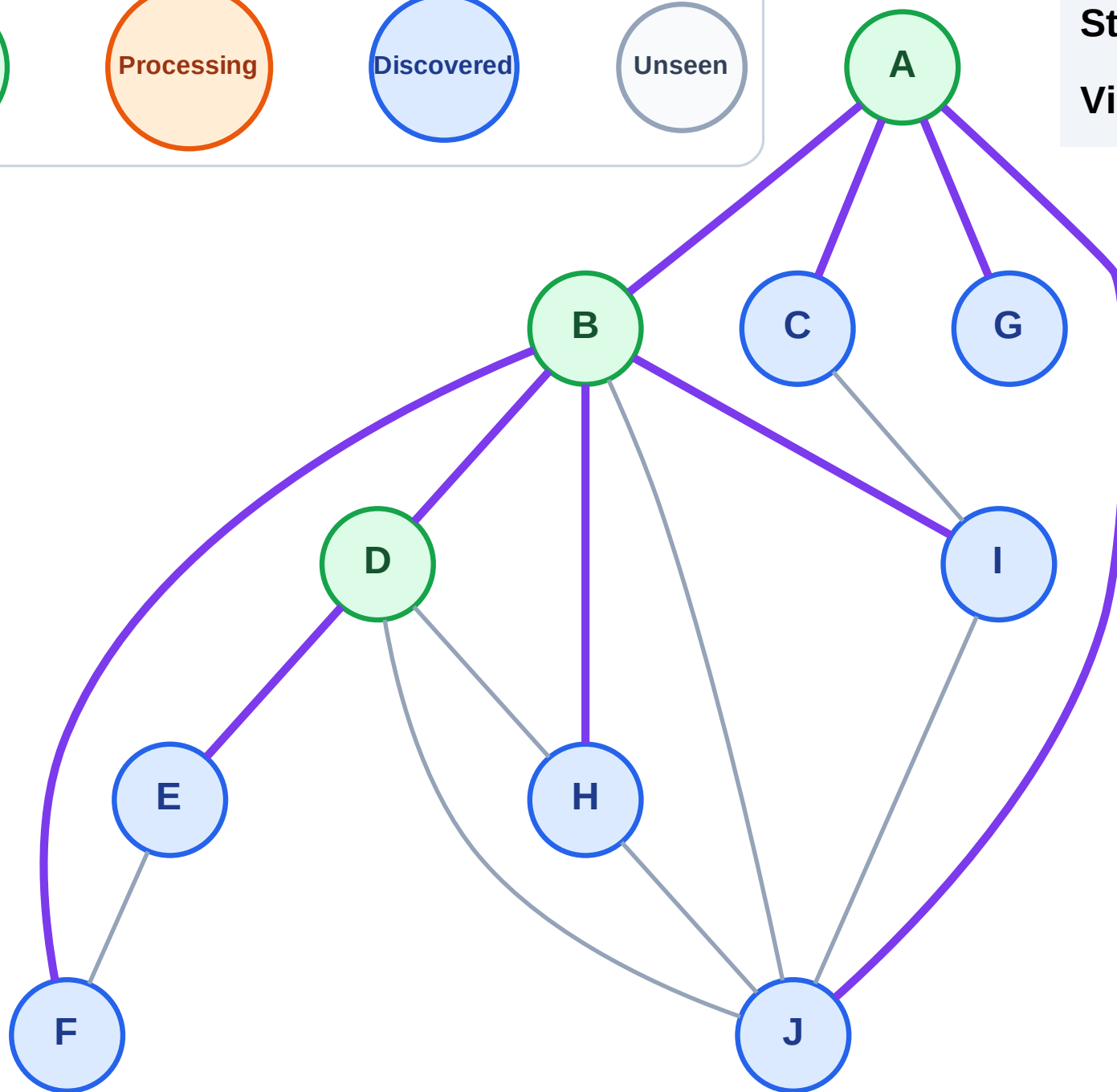
Step 7: Discover E from D

Legend

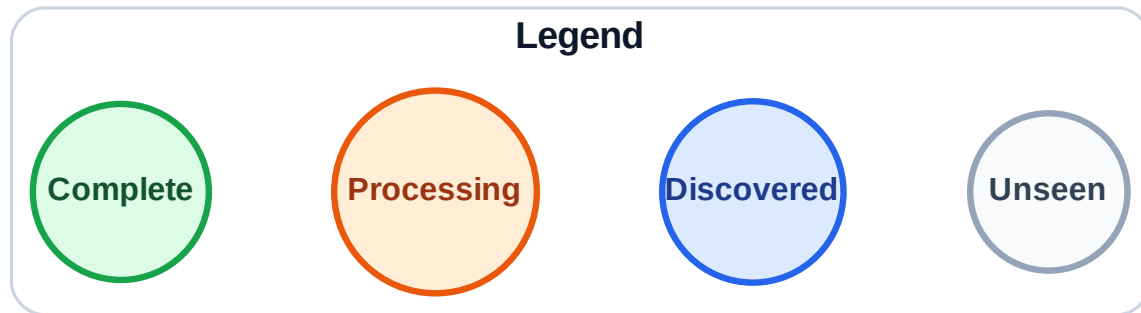


Stack (top at right): J | G | C | I | H | F | E

Visited: A, B, D

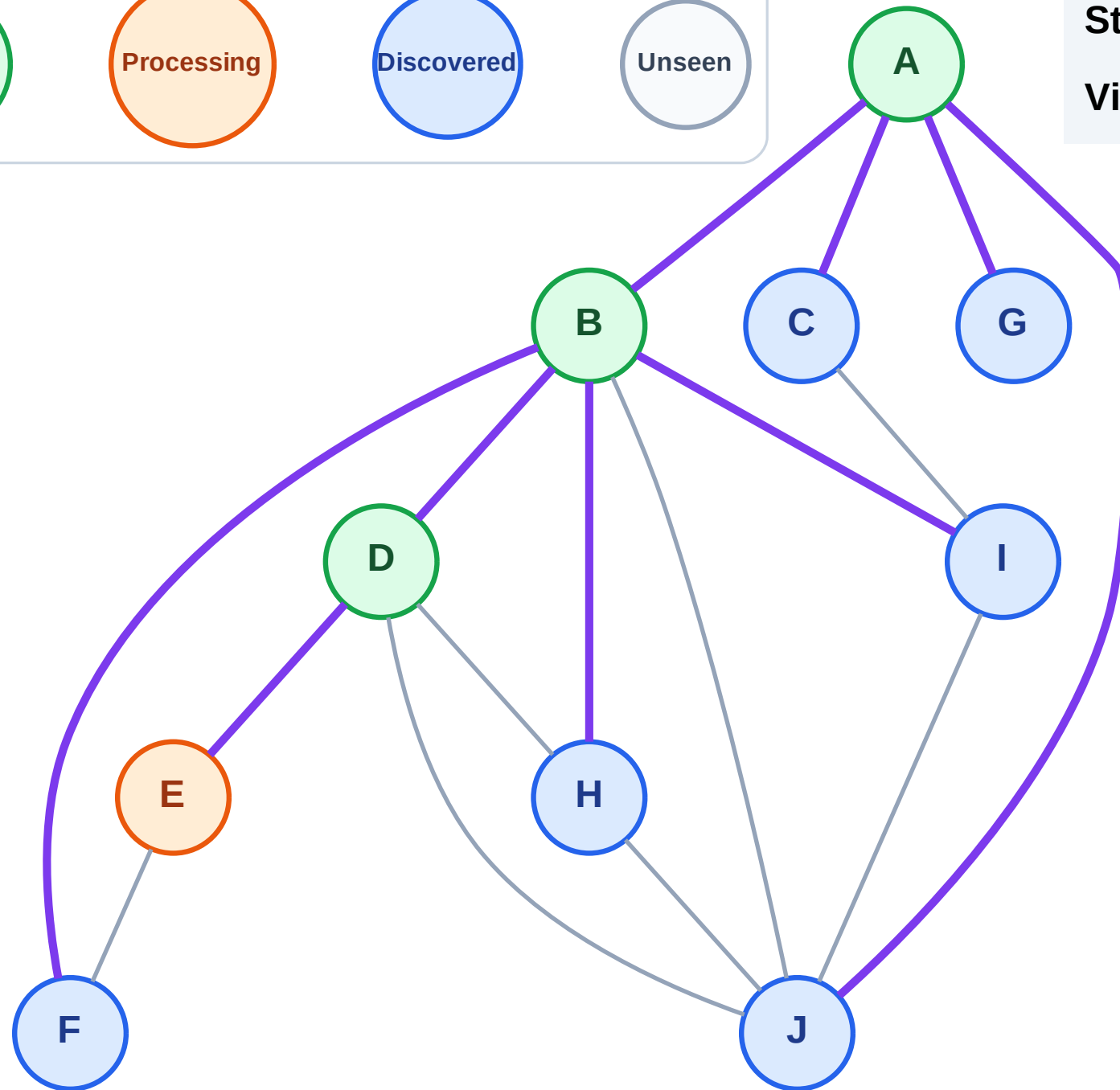


Step 8: Process node E



Stack (top at right): J | G | C | I | H | F

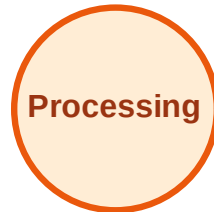
Visited: A, B, D



DFS Traversal

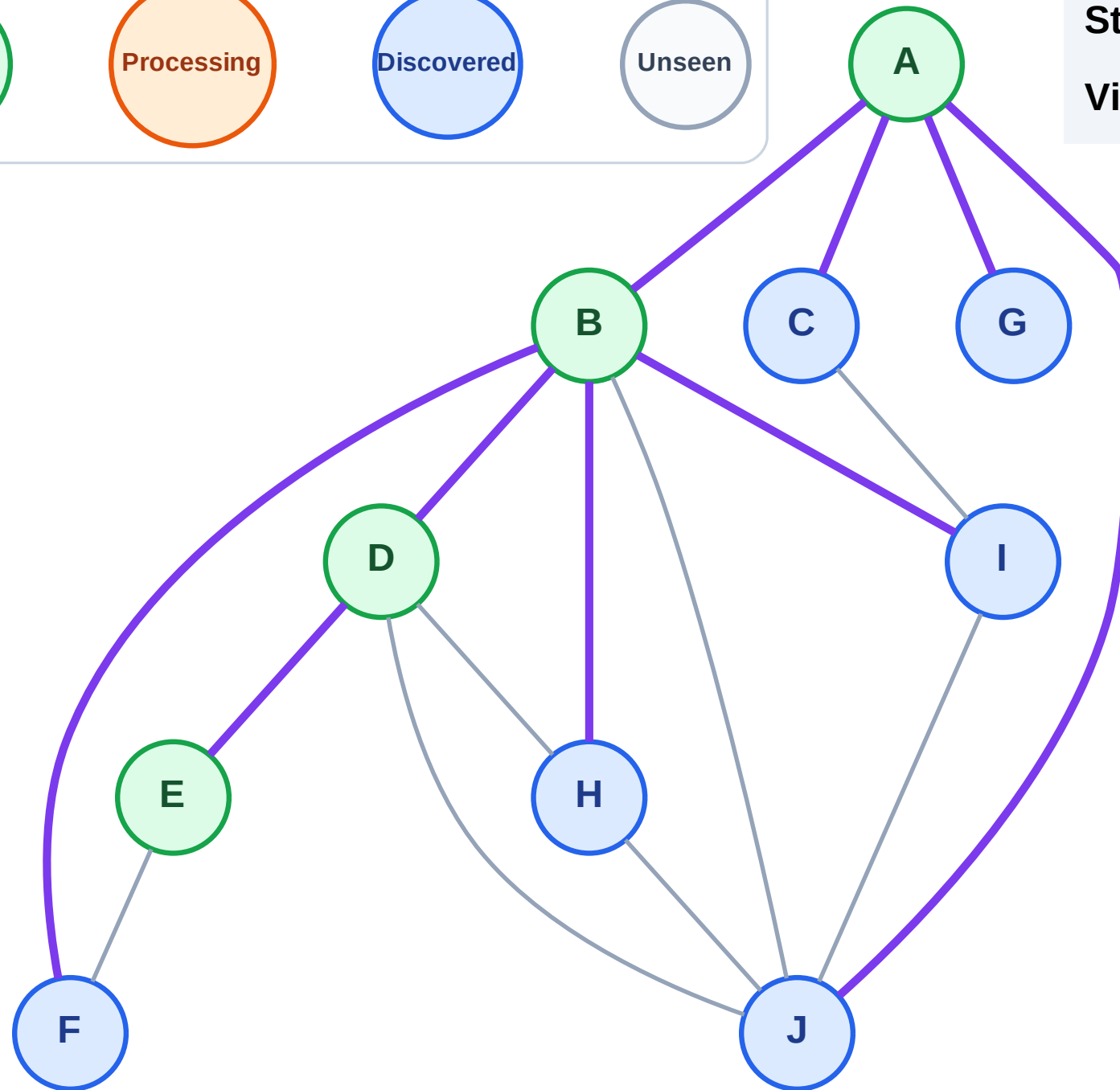
Step 9: No new neighbors from E

Legend



Stack (top at right): J | G | C | I | H | F

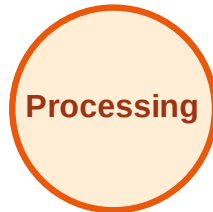
Visited: A, B, D, E



DFS Traversal

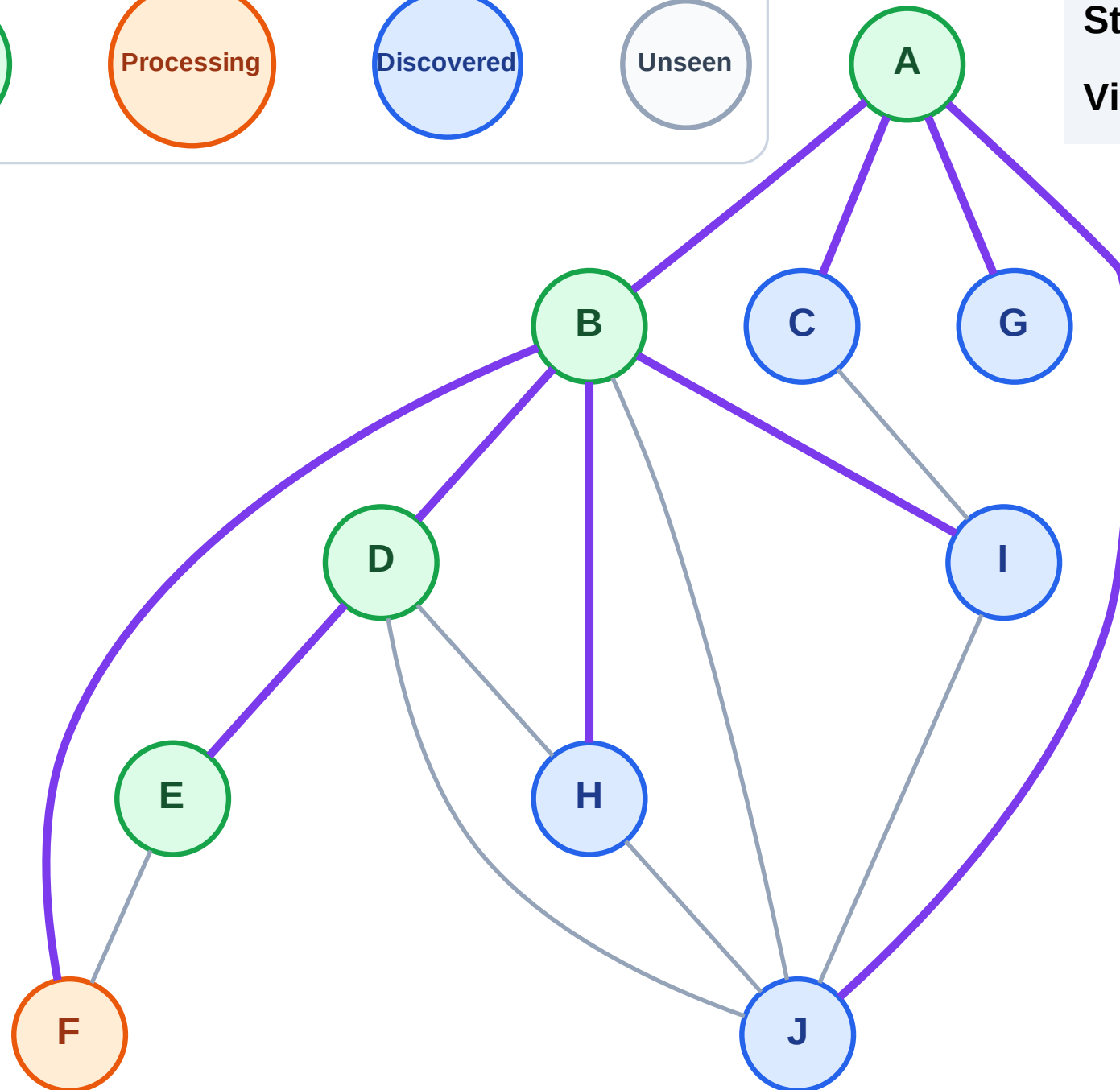
Step 10: Process node F

Legend



Stack (top at right): J | G | C | I | H

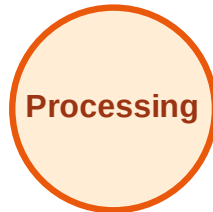
Visited: A, B, D, E



DFS Traversal

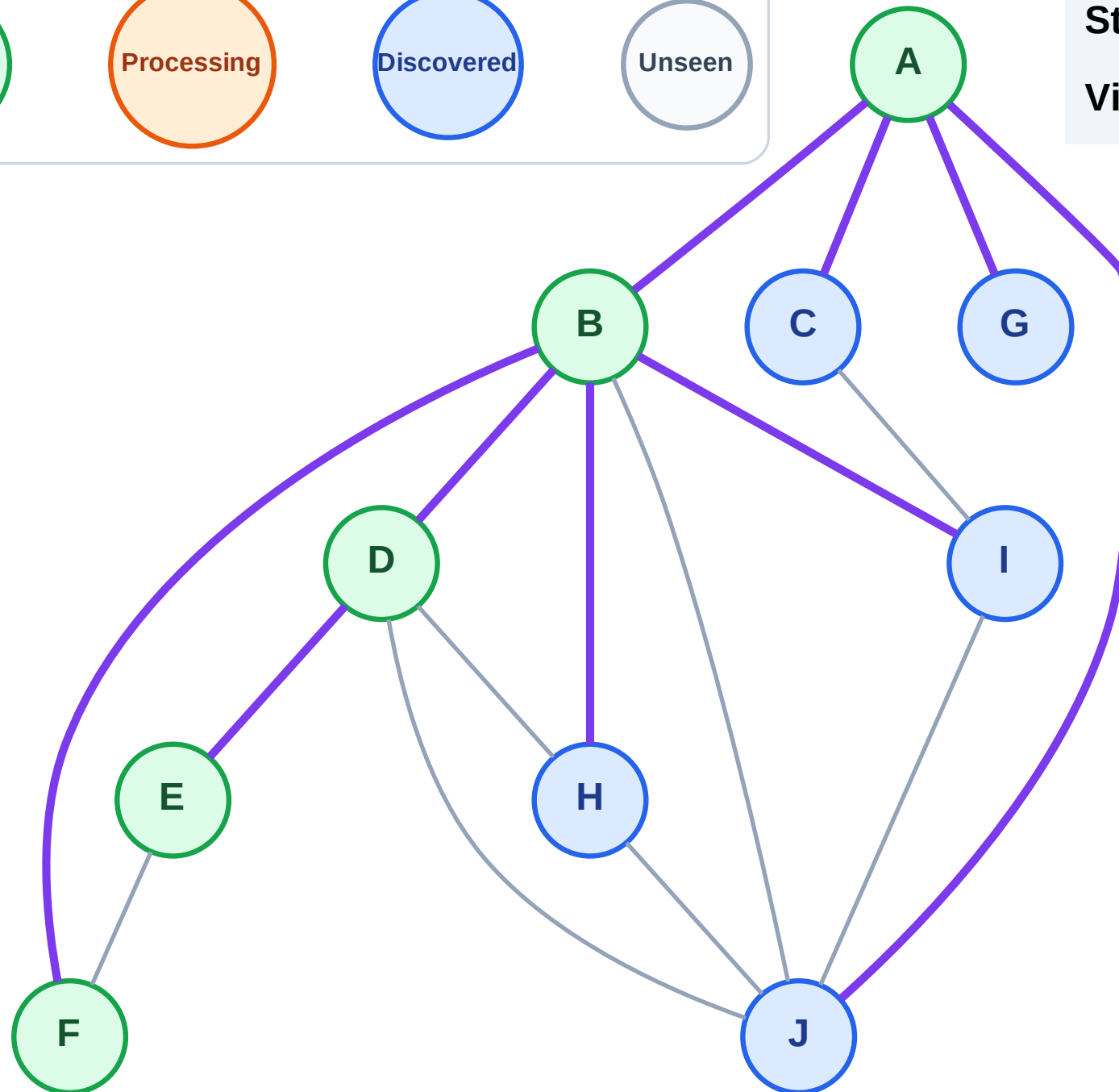
Step 11: No new neighbors from F

Legend



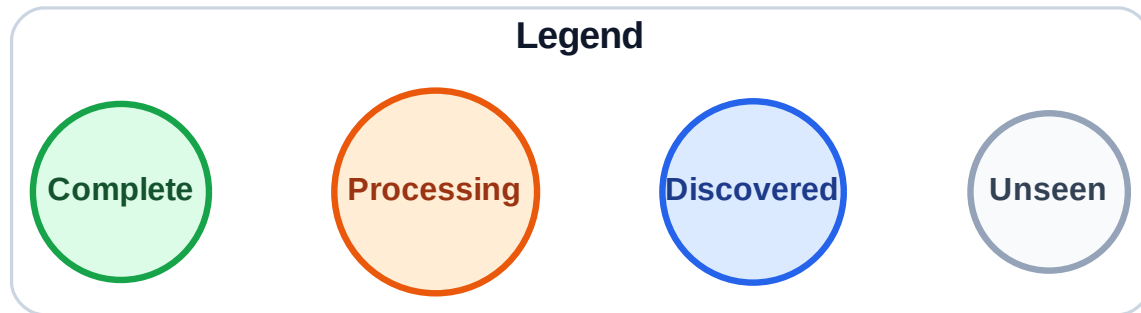
Stack (top at right): J | G | C | I | H

Visited: A, B, D, E, F



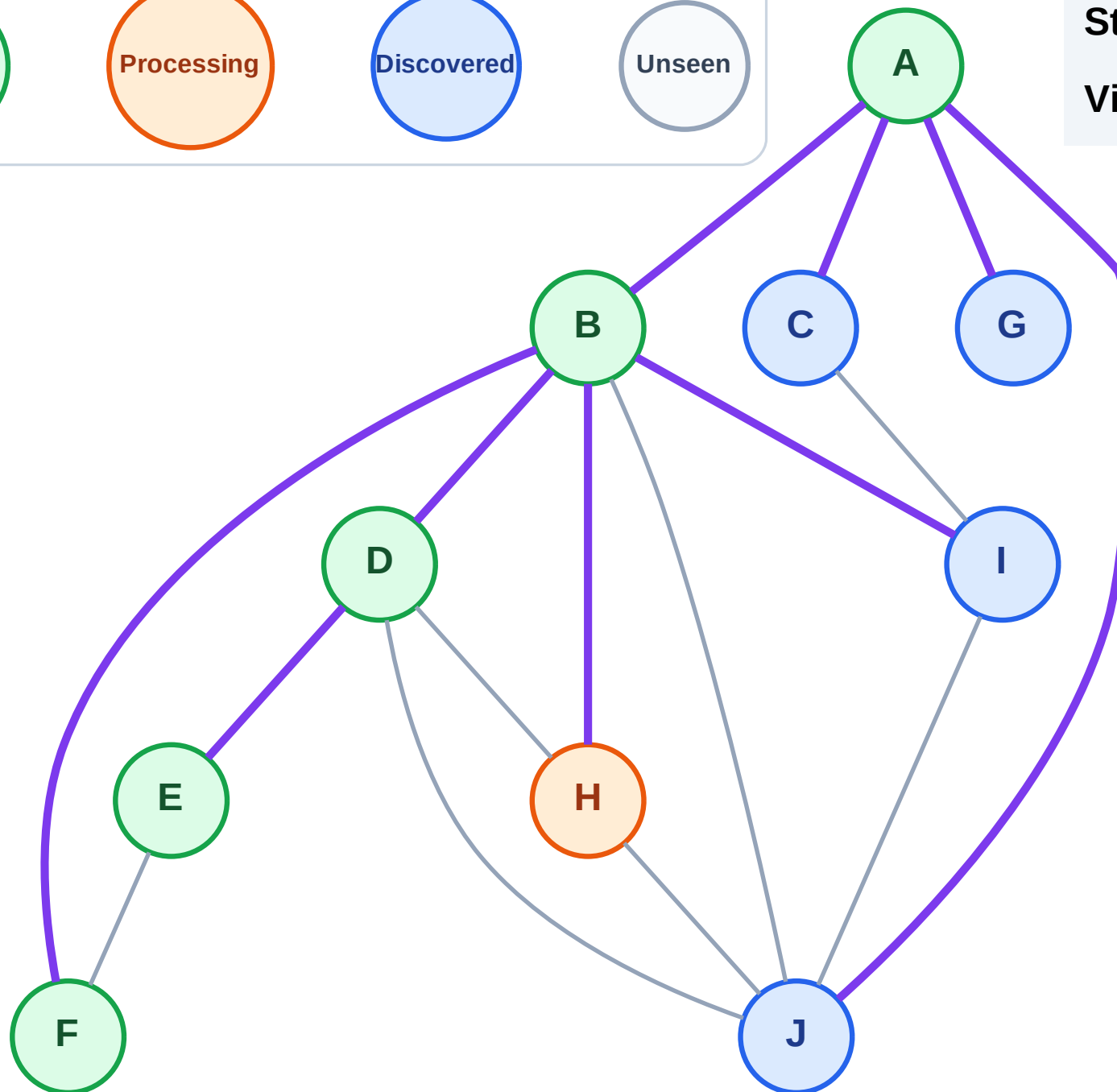
Step 12: Process node H

Step 12: Process node H



Stack (top at right): J | G | C | I

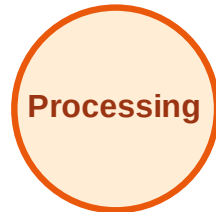
Visited: A, B, D, E, F



DFS Traversal

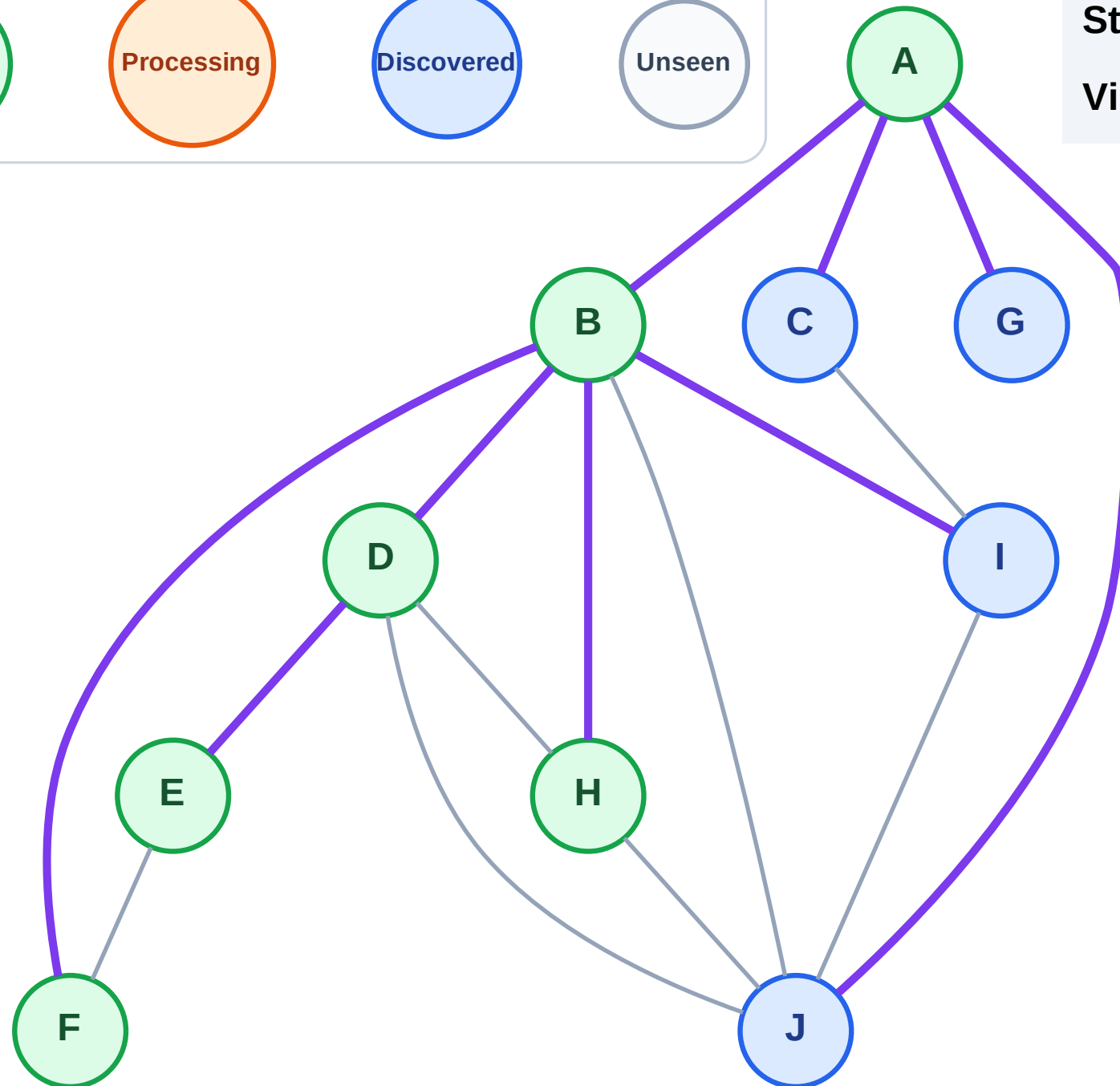
Step 13: No new neighbors from H

Legend



Stack (top at right): J | G | C | I

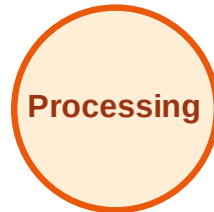
Visited: A, B, D, E, F, H



DFS Traversal

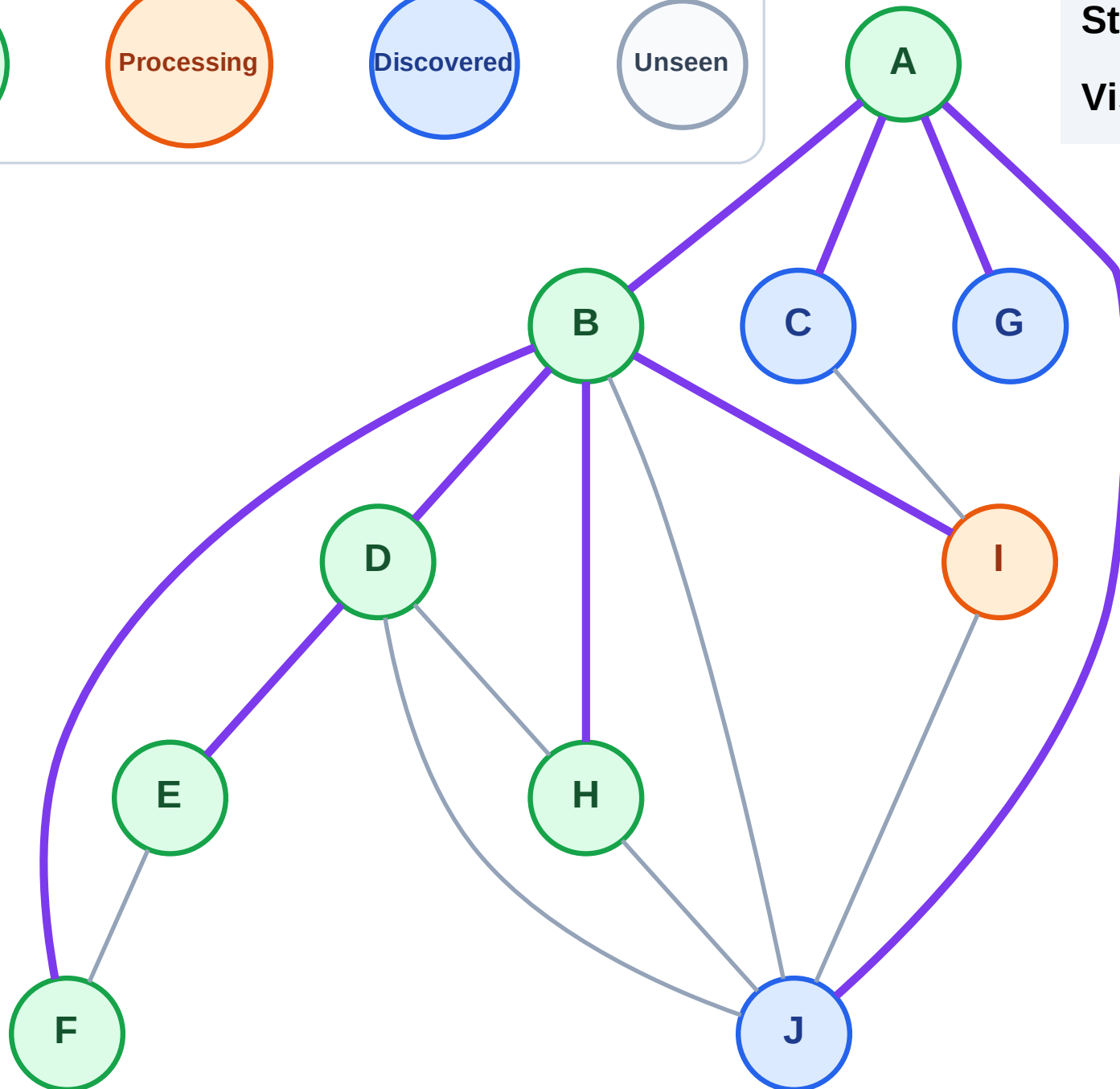
Step 14: Process node I

Legend



Stack (top at right): J | G | C

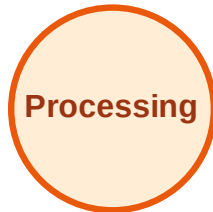
Visited: A, B, D, E, F, H



DFS Traversal

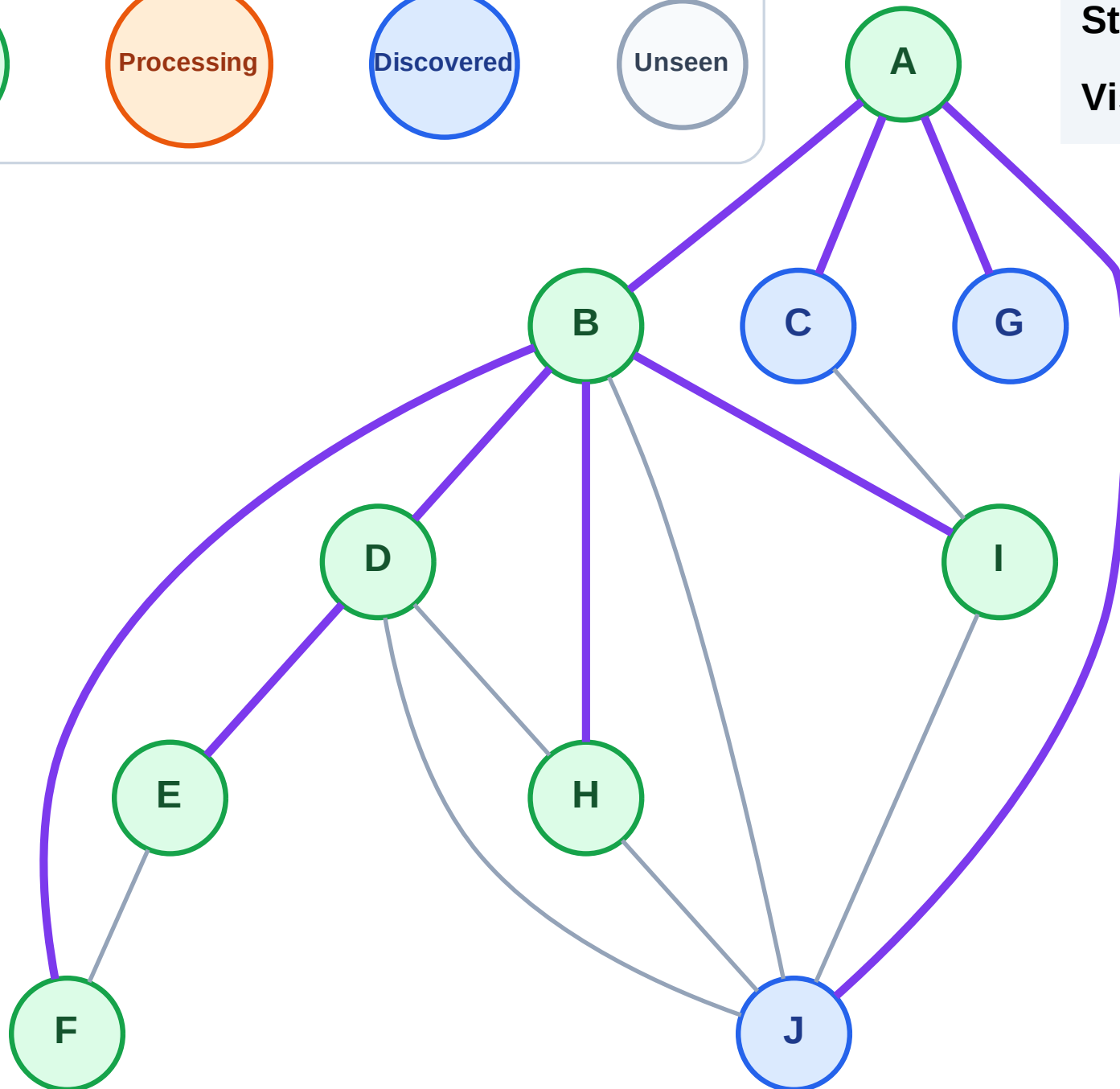
Step 15: No new neighbors from I

Legend



Stack (top at right): J | G | C

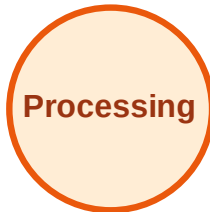
Visited: A, B, D, E, F, H, I



DFS Traversal

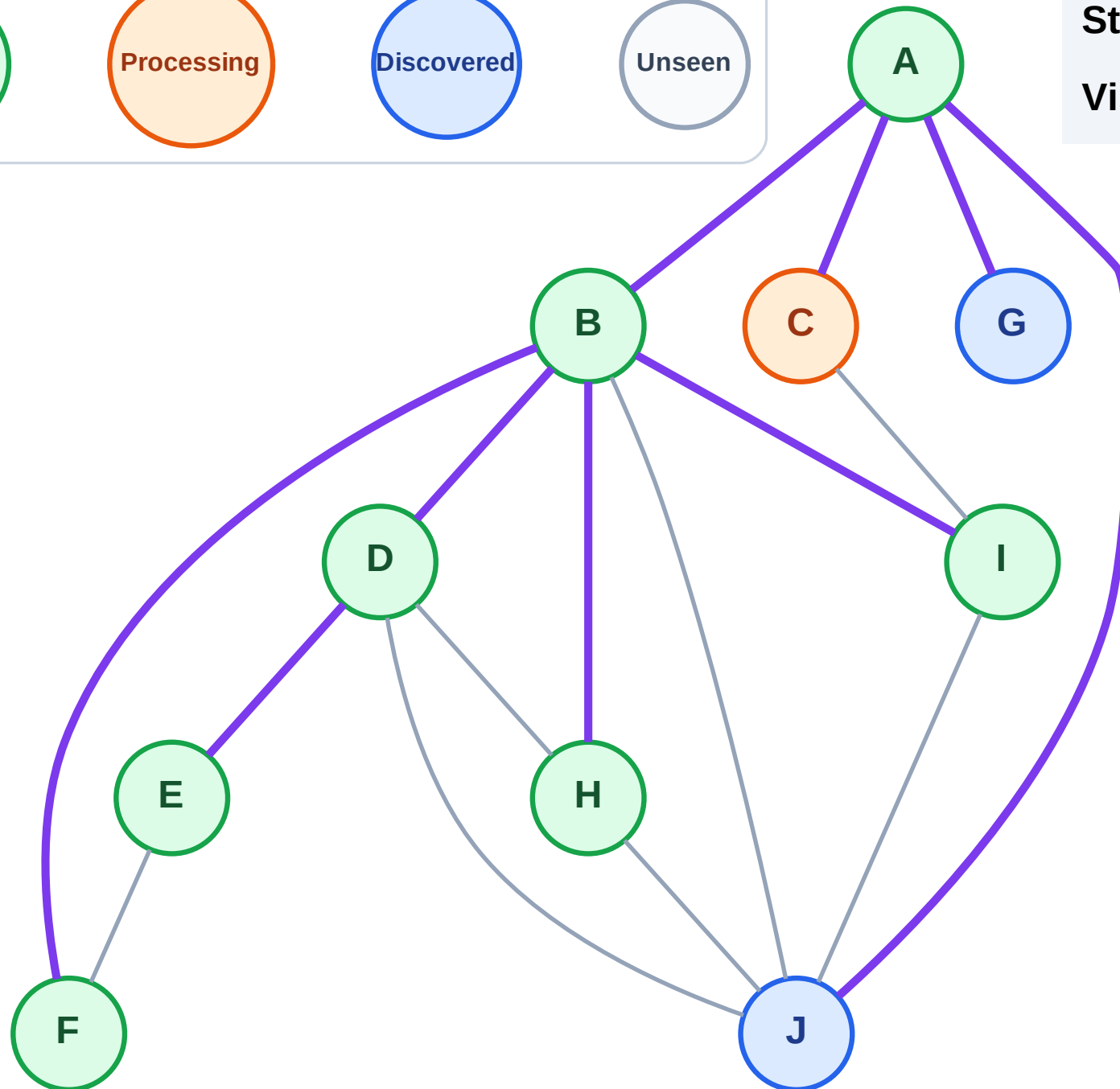
Step 16: Process node C

Legend



Stack (top at right): J | G

Visited: A, B, D, E, F, H, I



DFS Traversal

Step 17: No new neighbors from C

Legend

Complete

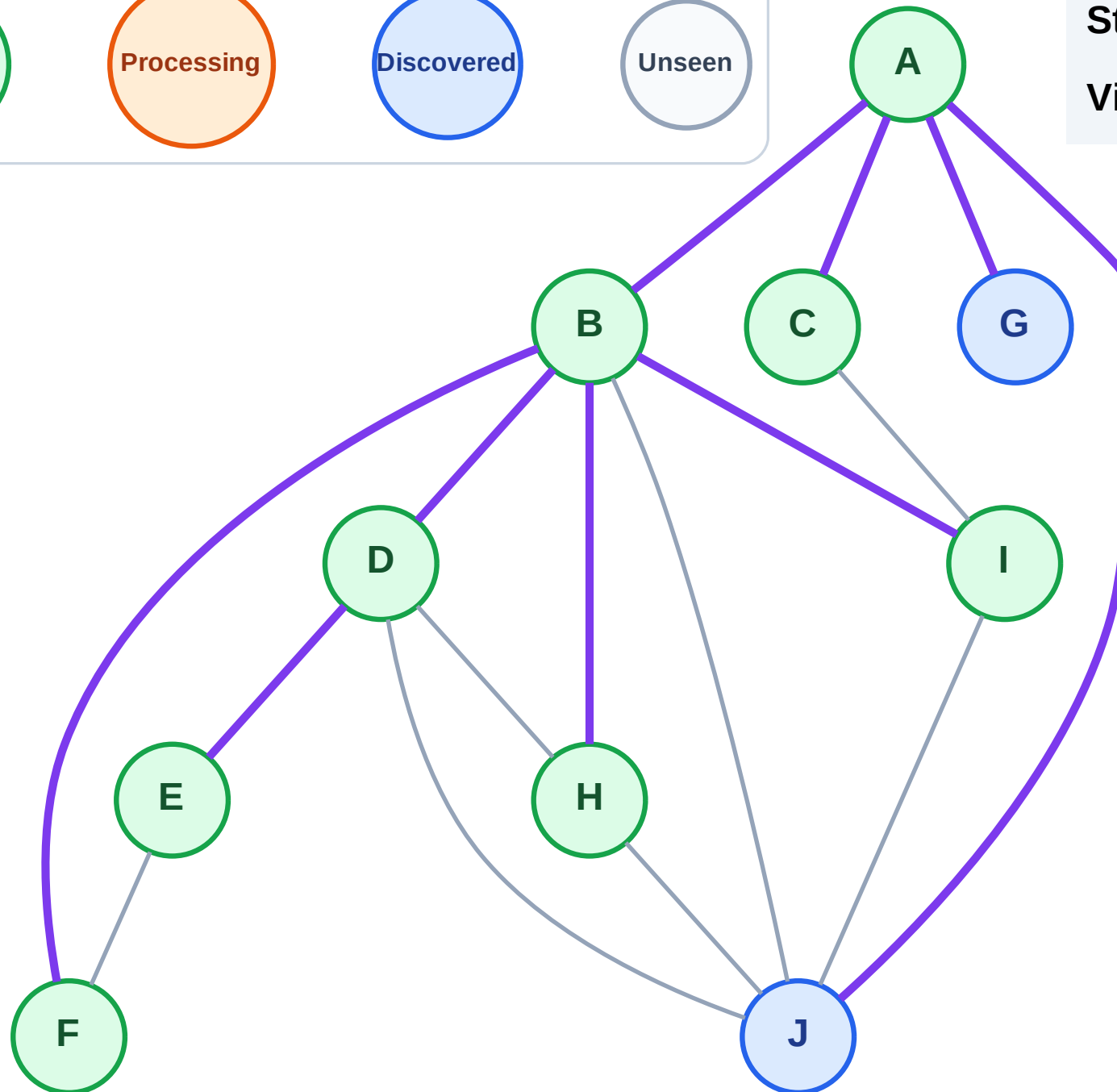
Processing

Discovered

Unseen

Stack (top at right): J | G

Visited: A, B, C, D, E, F, H, I



DFS Traversal

Step 18: Process node G

Legend

Complete

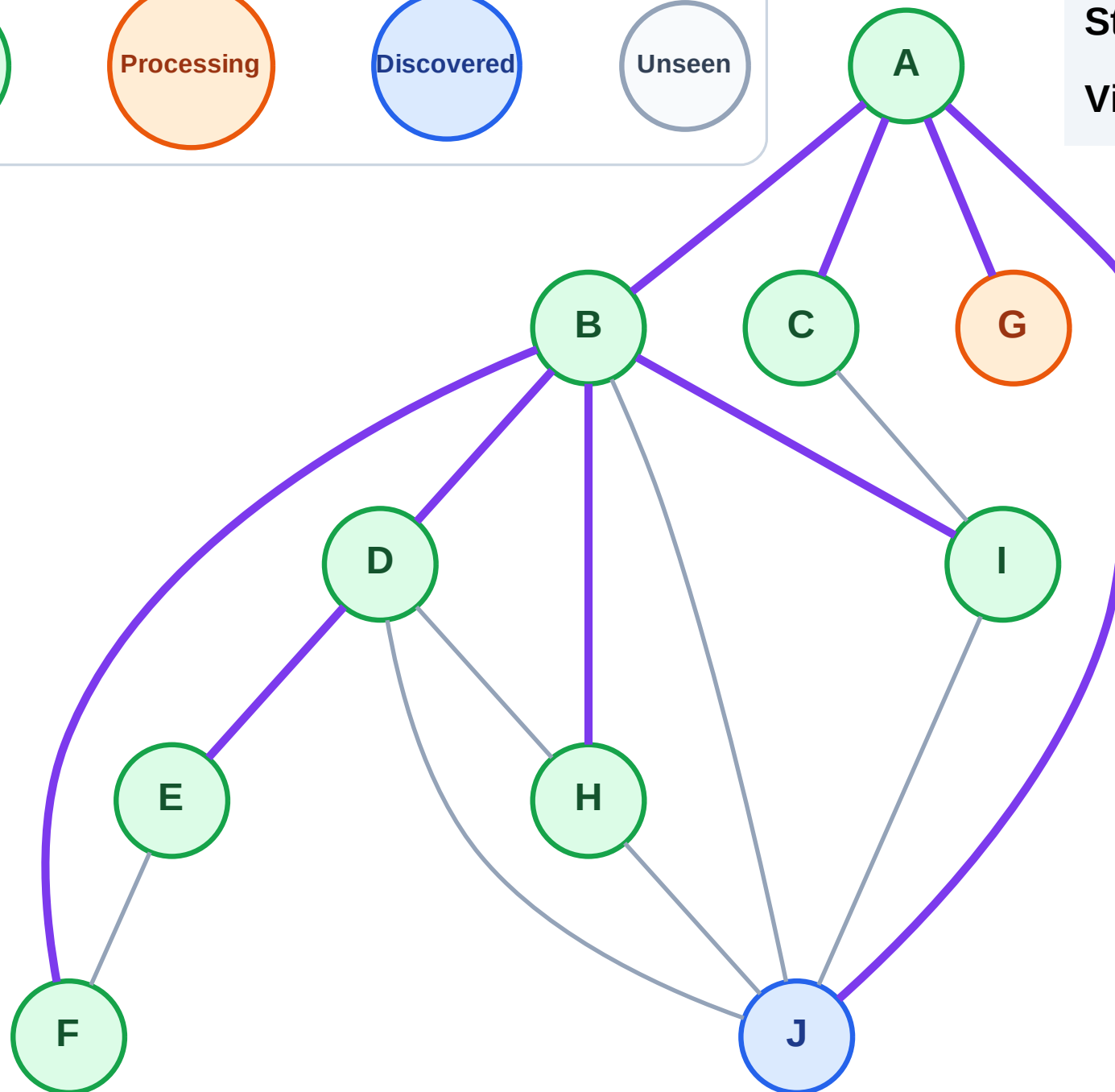
Processing

Discovered

Unseen

Stack (top at right): J

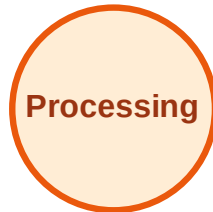
Visited: A, B, C, D, E, F, H, I



DFS Traversal

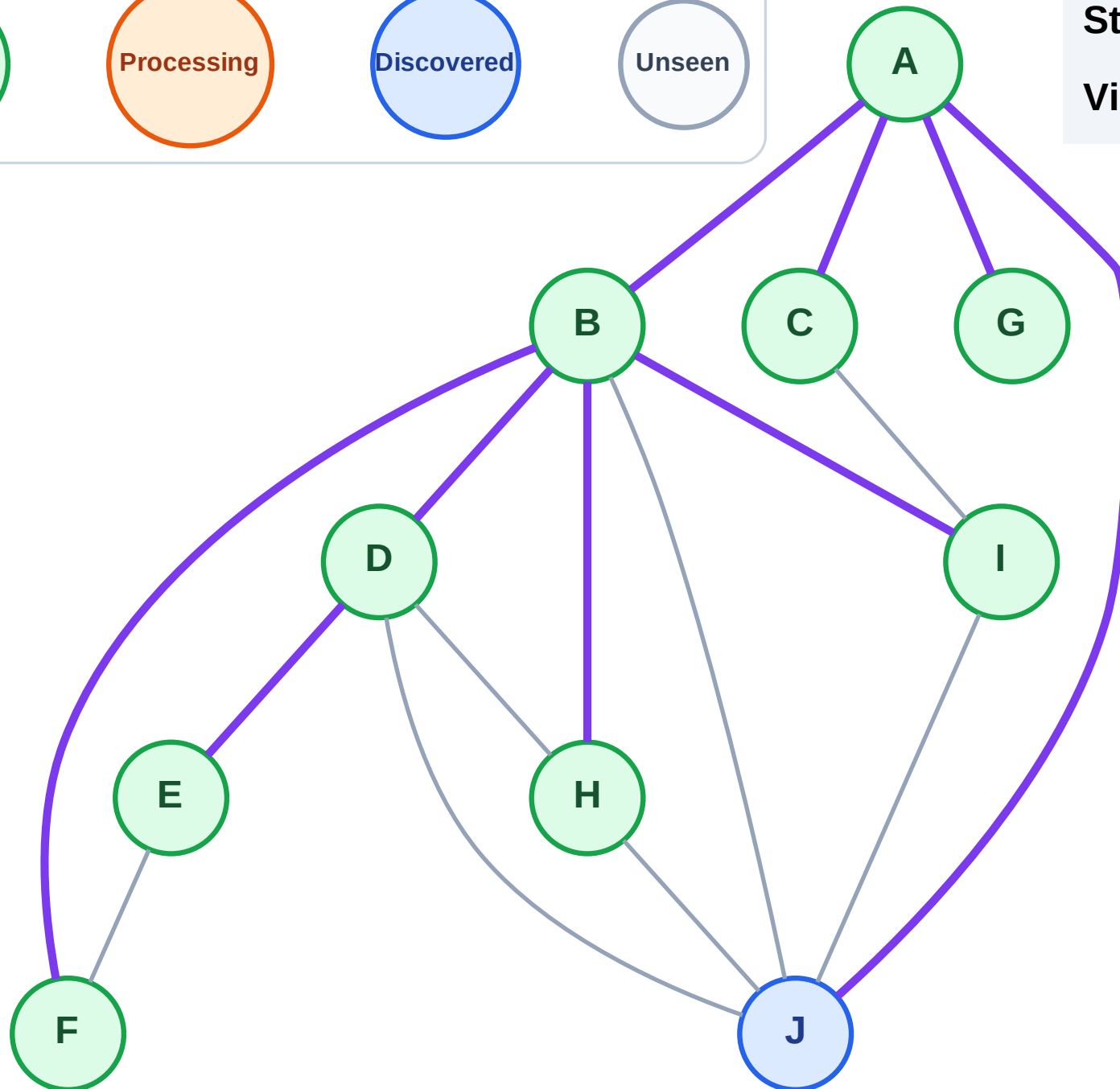
Step 19: No new neighbors from G

Legend



Stack (top at right): J

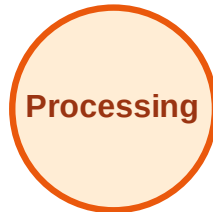
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

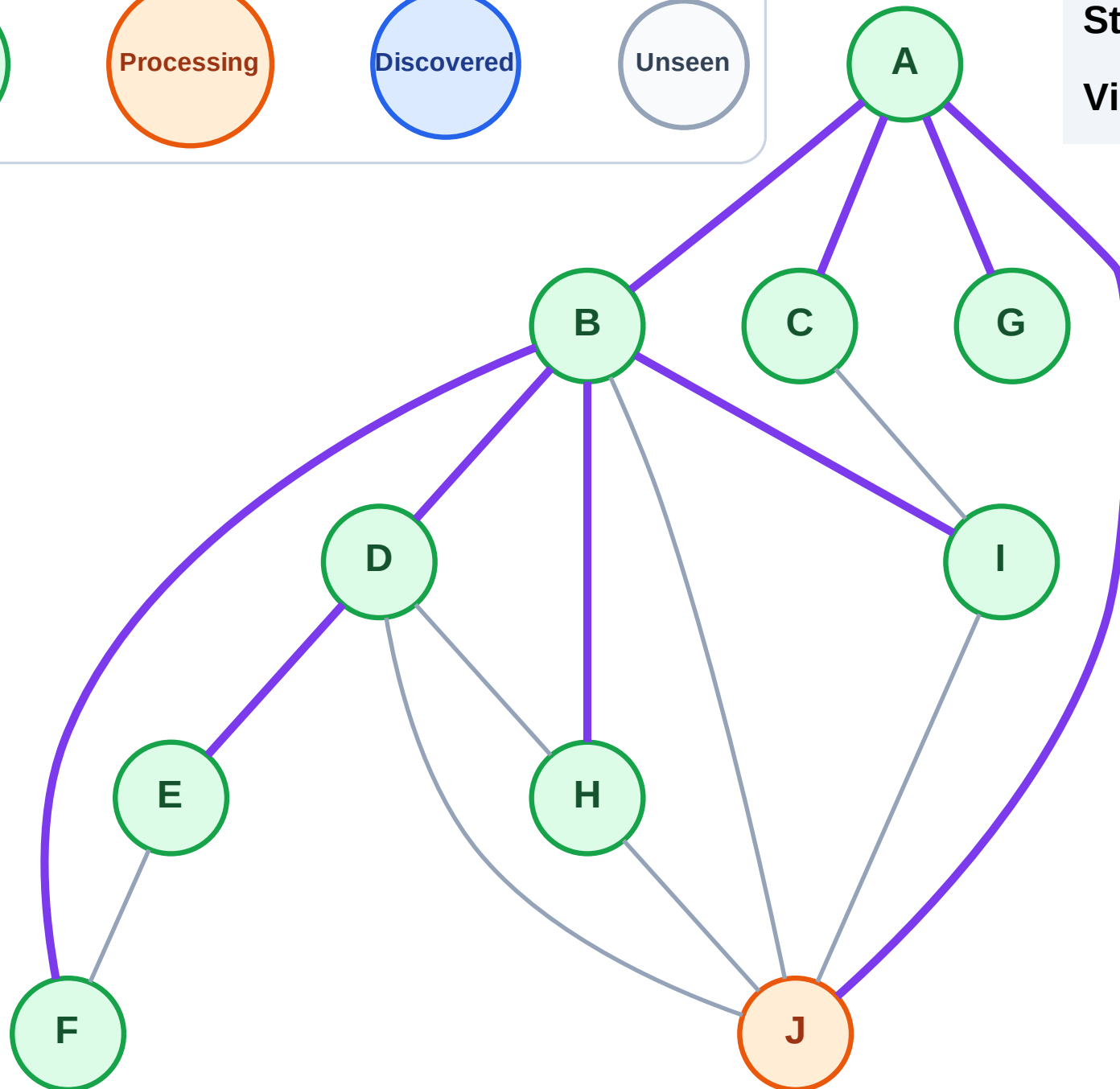
Step 20: Process node J

Legend



Stack (top at right): (empty)

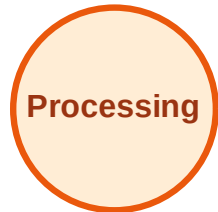
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

